PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

FOR

B.Sc. /B.Sc. (Hons)/B.Sc. (Hons) with research
Multidisciplinary Courses
(SEMESTER SYSTEM)

EXAMINATIONS, 2025-26
(SEMESTER: FIRST TO THIRD)

i.e

First &Third Semester : November/December, 2025
Second Semester : April/May, 2026

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Agriculture

Course Title: AGRICULTURAL BOTANY AND CROP PHYSIOLOGY **Course Code: (AGR-MDC-1-101)**
Semester: First

Learning Outcome: To learn ecological concepts and theories of evolutions, growth and development of crops.

Max. Marks: 75 **Course Credits** = 2+1=3

1.	Theory: 50	Written Exam: 45	Internal assessment: 5
2.	Practical: 25	Written Exam: 22	Internal assessment: 3

Periods per week: 1. Theory- Three of 45 minutes duration (Two Hours)
2. Practical- One of two hours duration (Two Hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 9 short answer type questions of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 9 marks each and one question is to be attempted out of each unit.

Teaching Aids: Teachers can use lecture, lecture cum demonstration, project, assignments, problem solving, inductive and deductive methods, and can use L.C.D., L.E.D., Multimedia projector in addition to black/white board.

UNIT – I — 8 Hours

Plant ecology, nature and scope of ecology, concept of environment.
Ecological factors and their influence on distribution of vegetation.
Main ecological plant communities and their adaptations.
Concept of ecosystem and biosphere.

UNIT – II — 7 Hours

Sources of pollution, environmental conservation and health.
Concept of center of origin of crop plants.
Domestication of crops and evaluation of germplasm.
Weed ecology and weed competitions.

UNIT – III — 8 Hours

Introductory concept of theories of organic evolution.
Evolution of important group of plants.
Variation, their types and causes, adaptations.
Growth and development of crops: Photosynthesis, respiration, net photosynthesis, translocation of assimilates.

UNIT – IV — 7 Hours

Growth of plants: development stages, physiological growth components, factors influencing growth and development.
Plant growth substances, growth expressions.
Transpiration, absorption of food material by plants.
Water absorption and translocation.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 Hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Growth analysis experiments demonstrating various aspects of respiration, photosynthesis, and ascent of sap.
2. Water culture techniques/ deficiency and toxicity of major and minor elements.

Suggested Readings:

1. Devlin, R.M., *Plant Physiology*, Prindle Weber &SvhmidtPublisher, New York, 1983.
2. Kochhar, P.L., *Plant Physiology*, Trumen Publishers Jalandhar, 2010.
3. Bhatia K.N., and Widge, R., Foundation *of Botany*, Truman Publishers, Jalandhar, 2010.

Course Title: AGRICULTURAL MICROBIOLOGY **Course Code: (AGR-MDC-2-201)**
Semester: Second

Learning Outcome : To learn various applications of micro-organism in agriculture, industry and environment.

Max. Marks: 75			**Course Credits = 2+1=3**
1.	**Theory: 50**	**Written Exam: 45**	**Internal assessment: 5**
2.	**Practical: 25**	**Written Exam: 22**	**Internal assessment: 3**

Periods per week: 1. Theory- Three of 45 minutes duration (Two Hours)
2. Practical- One of two hours duration (Two Hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 9 short answer type questions of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 9 marks each and one question is to be attempted out of each unit.

Teaching Aids: Teachers can use lecture, lecture cum demonstration, project, assignments, problem solving, inductive and deductive methods, and can use L.C.D., L.E.D., Multimedia projector in addition to black/white board.

UNIT – I **8 Hours**

Microbiology: Introduction, various branches of microbiology, history of soil microbiology.
Classification of soil organisms: animalia, plantae, fungi, Monera.
Soil bacteria classification, soil viruses, optimum conditions for microbial activity.
Functions of soil micro-organisms, role of microbes in soil fertility.

UNIT – II **7 Hours**

Different approaches to study soil microbiology
Food preservation methods: Physical and chemical preservation methods
Biological nitrogen fixation, legume, rhizobium symbiosis.
Vermi composting: types of vermicomposting, method of vermicomposting, nutrient content of vermicompost.

UNIT – III **8 Hours**

Environmental Microbiology: Microbiology of organic compost,
Bio-filtration, purification of water, sedimentation, filtration and disinfection
Microbiological analysis of water purity.
Types of soil microbes, organic matter decomposition, composition of litter, carbon assimilation and immobilization, microorganism associated with organic matter decomposition.

UNIT – IV **7 Hours**

Organic matter dynamics in soil, factor affecting organic matter decomposition.
Biogeochemical cycle of carbon, nitrogen, phosphorus and sulphur.
Crop protection using microorganisms: Microbial herbicides, bacterial insecticides.
Virus insecticides, entomopathogenic fungi, microbiology of lignocellulose degradation in rumen.

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Methods in Microbiology – cultivation and isolation of microorganisms
2. Demonstration of techniques for pure culture of microorganism by streek plate, pour plate and spread plate technique
3. Methods of culture preservation and maintenance.
4. Measurement of microbial activity in soil by respiration method.
5. Identification of symbiotic bacteroid of rhizobia.
6. Observation of root colonization by V.A.M. fungi.
7. Enzymatic test of milk by methylene blue reductase test
8. Demonstration of micro-organisms in soil, water, air and food.

Suggested Readings:

1. Dubey, R.C., and Maheshwari, D.K., *A text book of Microbiology,* S. Chand & Company Ltd, New Delhi, 2010.
2. Peleczar. M.J., Chain, E. and Kreis, N.R., *Microbiology*, Tata-Mcgraw-Hill Company, 2008.
3. Alcamo I.E., *FundamentalsofMicrobiology*, Benjamin Cimmings Publishing Company, 2007.
4. Rangaswami, G., Bagyaraj, D. J. and Bagyaraj, D.G., *Agricultural Microbiology*, Prentice Hall of India Ltd. New Delhi, 2005.

B.Sc. Agriculture(4 year course) Multi-Disciplinary

Course Title: Statistical Techniques in Agriculture **Course Code: (AGR-MDC-3-301)**
Semester: Third

Learning Outcome: To learn basic statistical techniques for analysis of agricultural data.

Max. Marks: 75			**Course Credits** = 2+1=3
1.	Theory: 50	Written Exam: 45	Internal assessment: 5
2.	Practical: 25	Written Exam: 22	Internal assessment: 3

Periods per week: 1. Theory- Three of 45 minutes duration (Two Hours)
2. Practical- One of two hours duration (Two Hours)

Instructions for Paper Setters:
Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 9 short answer type questions of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 9 marks each and one question is to be attempted out of each unit.

Teaching Aids: Teachers can use lecture, lecture cum demonstration, project, assignments, problem solving, inductive and deductive methods, and can use L.C.D., L.E.D., Multimedia projector in addition to black/white board.

UNIT – I **8 Hours**

Definition of statistics, scope and limitation.Introductory knowledge regarding concept of primary and secondary data.
Construction of frequency distribution tales for quantitative variables.
Diagrams and graphs (simple, multiple bar diagram, pie chart and histogram)
Measure of central tendency– AM, Mode, medium.

UNIT – II **7 Hours**

Measure of dispersion- Range, SD, CV.
Correlation analysis: scatter diagram, simple correlation coefficient, properties of correlation coefficient.
Simple regression analysis

UNIT – III **8 Hours**

Test of significance – Small sample test (t test for single sample mean, t test for difference between two sample means, paired t test,).
Large sample test- (Z test for single mean, Z test for difference between two means, Z test for difference of proportions). F test to test equality of variance, Test of goodness of fit.

UNIT – IV **7 Hours**

Design of experiments: Preliminaries,
Principle of experiment design.
Completely randomized design (CRD),
Randomized block design (RBD).

GUIDELINES FOR PRACTICAL EXAMINATION

Time: 3 hrs **Max Marks: 25**
Practical: 22 **Internal Assessment: 03**

1. Calculation of Mean, Median, Mode, SD, CV
2. Correlation and regression analysis
3. Analysis of CRD, RBD

Suggested Readings:

Panse, V.G., Shaw, F.J., and Sukhatme, P.V., *Statistical methods for agricultural workers*, Indian Council of Agricultural Research, 1967.
Fisher, R.A., *Statistical methods for research workers (14th Edition),*Hafner Press, UK, 1975.
Singh, S., Singh, T.P., Bansal, M.L., and Kumar R., *Statistical Method for Research workers*Kalyani Publishers, Ludhiana, 2004.

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Biochemistry

Multi-Disciplinary/Inter-Disciplinary Paper

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 8 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-MDC-1-101): Introduction to Biochemistry (2+1)

Marks: 50 (40 +10 Int.)

Objective: To learn about biomolecules, their structure and function in the biological system.

Learning Outcomes:

- To understand the chemical and molecular foundations of life and their roles in biological systems
- To comprehend the classification, structure, chemical properties and biological functions of carbohydrates, lipids,proteins and nucleic acids
- To perform independently the analysis of various biomolecules quantitatively in the laboratory.

SECTION – I

(Lectures: 8)

Water: Physico chemical properties of water, dissociation and association constants. pH and buffers, pI, pka, Henderson Hasselbatch equation and its implications. **Carbohydrates**: Structure of important mono, di-, oligo- and polysaccharides, glycoproteins, peptidoglycan, glycolipids and lipopolysaccharides. Storage and cell wall polysaccharides.Enantiomers and anomers.Acid mucopolysaccharides.

SECTION- II

(Lectures: 8)

Lipids: Classification of lipids and fatty acids, general functions and structure of major lipid subclasses, acylglycerols, phosphoglycerols, phosphoglycerides, sphingolipids, glycosphingolipids and terpenes, sterols,steroids: Prostaglandins.

SECTION – III

(Lectures: 8)

Proteins: Structure of amino acids, nonprotein and rare amino acids and their chemical reactions. Structural organization of proteins (primary, secondary, tertiary and quaternary domain structure), protein classification and function.Forces stabilizing primary, secondary and tertiary structure.

SECTION-IV

(Lectures: 8)

Nucleic Acids: Structure and properties of purine and pyrimidine bases. Nucleosides and nucleotides.Structures of RNA & DNA.Porphyrins: General introduction

Suggested Books:

1. Lippincott's Illustrated Reviews: Biochemistry by P.C. Champe, R.A. Harvey & amp; D.R. Ferrier, Indian Edition, 2017.
2. Harper's Illustrated Biochemistry by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 31th Ed. 2018, Prentice Hall International Ltd. (U.K.).
3. Voet D and Voet JG. Biochemistry 5 th Edition, John Wiley and Sons Inc., 2021.
4. Lehninger A.L., Nelson D.L. and Cox M.M.(2021)8th edition Principles of biochemistry (W.H. Freeman, USA)
5. StryerL., J.M. Berg, J.L. Tymoczko (2023)10th edition, Biochemistry (W.H. Freeman Company, New York)
6. Conn E.E. and Stumpf P.K. (2006)5th edition Outlines of Biochemistry, (John Wiley and Sons, USA)
7. R.H. Garett and C.M. Grisham (2023)7th Edition Biochemistry, (Cengage Learning, US)
8. Zubay G.L., Parson W.W, Dennis E.Vance (William C.Brown, U.S.)1998Principles of biochemistry volume-1

PRACTICALS:

(BCH-MDC-1-101): Introduction to Biochemistry

Marks: 25 (20 +5 Int.)

1. Preparation of biological buffers and solutions of different pH and molarity.
2. Qualitative tests for carbohydrates.
3. Qualitative tests for amino acids.
4. Estimation of saponification value of fat/oil.
5. Estimation of acid value of fat/oil.
6. Titration curve of an amino acid and determination of its pKa and isoelectric pH.
7. Isolation of casein from milk.
8. Isolation of starch from potato/gluten from wheat.
9. Estimation of vitamin C

Multi-Disciplinary/Inter-Disciplinary Paper

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 8 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-MDC-2-201): Fundamentals of Enzymology (2+1)

Marks 50 (40 +10 Int.)

Objective: To learn principles of thermodynamics, nature of enzymes, their mode of action and enzyme kinetics.

Learning Outcomes:

- Will be able to learn the types of enzymes, their classification and their importance.
- Enzyme kinetics and enzyme inhibitors and the mechanisms of enzyme action
- Application of enzymes in diagnostics and various industries.
- Independently perform experiments of enzyme assay and enzyme activity.

SECTION-I

(Lectures: 8)

Introduction to enzymes.General characteristics of enzymes.IUB system of nomenclature and classification of enzymes.Prosthetic group.Holoenzyme, apoenzyme, cofactors, isoenzymes and their clinical significance and zymogens, coenzymes (NAD/NADP, FMN/FAD, coenzyme A, TPP, pyridoxal phosphate, tetrahydrofolate) and their biochemical functions, assay of enzyme activity, units of enzyme activity. Active site of enzymes.

SECTION-II

(Lectures: 8)

Enzymes as catalysts. Theories of enzymes catalysis: proximity and orientation effects, acid base catalysis, covalent catalysis. Role of metals in enzyme catalysis Monomeric enzymes, oligomeric enzymes &multienzyme complexes.Mechanism of action of chymotrypsin and pyruvate dehydrogenase and Regulation of enzyme activity.

SECTION-III

(Lectures: 8)

Factors affecting velocity of enzyme catalyzed reactions: enzyme concentration, pH and temperature.Michaelis-Menten equation.Determination of Km and its significance.
Enzyme inhibition: Various types of enzyme inhibitions. Determination of Ki value.Enzyme inhibitors andtheir importance.Introduction to multisubstrate enzyme, Allosteric enzymes properties and regulation.

SECTION-IV

(Lectures: 8)

Purifications of enzymes.Abzymes, Industrial Applications of enzymes: applications of enzymes in food, leather, clinical and biotechnology industry.

Suggested Books:

1. T. Palmer Enzymes: Biochemistry, Biotechnology and Clinical Chemistry, 2nd Ed. (2018) Affiliated East-West press Pvt. Ltd. New Delhi.
2. L. Stevens and Nicholas C.Price Fundamentals of Enzymology: The cell and molecular Biology of catalytic Proteins (2000) 3rd edition (Oxford University Press, USA).
3. Robert A. Copeland: A practical Introduction to Structure, Mechanism and Data Analysis2nd edition (John Wiley and Sons Inc. Publications).
4. Lehninger A. L., Nelson D. L. and Cox M. M. (2021) 8th edition Principles of biochemistry (W. H. Freeman, USA).
5. Stryer L., J. M. Berg, J. L. Tymoczko (2023) 10th edition, Biochemistry (W. H. Freeman company, New York) 6. Conn E. E. and Stumpf P. K. (2006) 5th edition Outlines of Biochemistry, (John Wiley and Sons, USA).
6. Kennely Peter. J and Kathleen M. Botham, IE Harpers illustrated biochemistry 32nd edition 2023 (McGraw-Hill Education).

PRACTICALS:

(BCH-MDC-2-201): Fundamentals of Enzymology

Marks: 25 (20 +5 Int.)

1. Determination of amylase activity.
2. Assay of alkaline phosphatase/Acid Phosphatase activity.
3. Effect of pH on Acid phosphatase activity.
4. Effect of temperature on enzyme activity and determination of energy of activation.
5. Effect of substrate concentration on enzyme activity and determination of Km.
6. Effect of Inhibitors on enzyme Activity.
7. Demonstration of potato polyphenol oxidase activity

Multi-Disciplinary/Inter-Disciplinary Paper

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:

1. Examiner will set total of NINE questions comprising TWO questions from each unit and ONE compulsory question.
2. All questions carry equal marks.
3. The compulsory question will consists of 8 short questions of one mark each covering the whole syllabi.
4. Besides Question No. 1, all other questions may contains 2-3 sub-parts.
5. Questions should be uniformly spread over the entire syllabus.
6. The students are required to attempt FIVE questions in all, ONE question from each unit and the Compulsory question.

(BCH-MDC-3-301): Basic Cell Biology (2+1) **Marks: 50 (40 +10 Int.)**

Objective: This course is designed to help students understand basis of the cell structure and function at cellular and molecular level.

Learning Outcomes:
Upon successful completion of this course, students are able to:

- Understand the structure and functions of the basic components of prokaryotic and eukaryotic cells, an Comprehend the structure and functions of chromosomes and phases of cell division.
- Independently gain practical hands-on expertise in basic microscopy, simple staining and isolation and different organelles.

SECTION – I

(Lectures: 8)

Cell as a unit of Life: PPLO's, bacteria, eukaryotic, microbes, plant and animal cells., The cell theory, Bio-membranes; structure and function, fluid mosaic concept, Faces of the membranes. Cell wall-structure and functions.Biochemical composition of cells (proteins, lipids, carbohydrates, nucleic acids and the metabolic pool.

SECTION- II

(Lectures: 8)

Cell Organelles: Non-membranous organelles: Structure and functions of ribosomes, centrioles and basal bodies Single membrane bound organelles: endoplasmic reticulum, golgi bodies and lysosomes, peroxisomes and glyoxysomes. Double membrane bound organelles: Mitochondria; structure and functions, semi-autonomous nature; endosymbiont hypothesis; mitochondrial DNA, Nucleus: Nuclear Envelope- structure of interphase nucleus; chromatin material, euchromatin and heterochromatin, nucleolus.

SECTION– III

(Lectures: 8)

Cell cycle: Overview of Cell cycle and its regulation, mitosis, meiosis, binary fission, amitosis, Cell senescence and death: Apoptosis and necrosis, Cell differentiation in plants and animals: totipotent, multipotent, pleuripotent cell.

SECTION-IV

(Lectures: 8)

Techniques in cell biology: Principles and applications of Light microscopy, Phase contrast microscopy, Confocal microscopy, Electron microscopy (SEM, TEM, STEM), fluorescence microscopy, Basics and uses of flow cytometry, fluorescent probes, Centrifugation :differential and density gradient, Chromatography: Paper and TLC.

Suggested Books:

1. Cell Biology by Thomas D. Pollard, William C. Earnshaw.
2. Cell Biology: Structure, Biochemistry, and Function by Phillip Sheeler, Donald E. Bianchi.
3. Molecular Biology of the Cell by Bruce Albert and Dennis Bray, 4th Ed. Garland Publishing Inc, New York and London.
4. Cell and Molecular Biology by Gerald Karp.1996. John Willey and Sons,Inc. London.
5. Cell and Molecular Biology- De-Robertis, F.D.P. and De-Robertis Jr. E.M.F. 6. Molecular Cell Biology- Lodish, H., Baltimore, D., Berk, A., Zipursky,

PRACTICALS:

(BCH-MDC-3-301): Basic Cell Biology

Marks: 25 (20 +5 Int.)

1. Microscopy: a. Use and care of Light compound microscope.
2. Study of structure of any Prokaryotic and Eukaryotic cell.
3. Isolation of chloroplasts by differential centrifugation.
4. To separate chloroplast pigments by chromatography.
5. Study of plasmolysis and de-plasmolysis.
6. Cell fractionation and determination of enzyme activity in organelles using sprouted seed or any other suitable source.
7. Preparation of Nuclear, Mitochondrial & cytoplasmic fractions.

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Bioinformatics

Semester – I

BINF-MDC-1-101: INTRODUCTION TO BIOINFORMATICS - I

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

External Theory Marks: 40
Internal Theory Marks: 10
Total Theory Marks: 50
Exam Duration:3 Hours

Course Duration:

1 Semester = 15 weeks which will have 30 hours of Theory Lectures (2 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 3 (2 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

Introduction to Bioinformatics exposes the students to study the basicsand applications of Bioinformatics.

General Instruction:

- The question paper will have seven questions, and each question have 08 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Introduction to Bioinformatics: Definitions, History, Goal, Scope and its Applications.

Literature Database: PubMed, PMC, NCBI Book Shelf.

Biological Databases Classification: Primary, Secondary and Specialized databases.
Nucleotide Sequence Database: Genbank, EMBL, DDBJ, INSDC, IBDC and its data portals.
Protein Sequence Database: Uniprot.
Database flat file formats: Genbank/Genpept, EMBL and Uniprot flat files.
Sequence file formats: FASTA and PIR Format.
Information retrieval for biological databases: Gquery, Boolean search terms and statements

UNIT – II

Sequence Alignment:

Evolutionary basis of Alignment, Concept of Sequence Homology, Similarity, and Identity.

Pairwise sequence alignment: Concept of Global and Local alignment, Dot matrix method, Dynamic programming (Needleman Wunsch algorithm and Smith-Waterman algorithm) Concepts behind General gap and affine gap penalty.

Substitution matrices: Amino acid substitution matrices (PAM and BLOSUM)

Tools for Sequence Alignment:BLAST: Algorithm, types, understanding the BLAST output and the significance E-value and Bit score, different uses of BLAST.

FASTA algorithm, Difference between BLAST & FASTA.

Applications of Sequence Alignments.

Recommended Books:

- Stallings (2006), Operating Systems, Internals and Design Principles, 5th edition,Pearson Education, India.
- Andrew S. Tanenbaum (2007), Modern Operating Systems, 2nd edition, Prentice Hall of India, India.
- Introduction to Bioinformatics by V. Kothekar, 1st Edition, (2006), Dhruv Publications, Delhi.
- Bioinformatics: A modern approach by Vittal R. Srinivas, Prentice/ Hall of India Pvt. Ltd. New Delhi, (2005)
- Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
- Bioinformartdics: A practical guide to the analysis of genes and proteins. Ed. By
- Baxevanis, 3rd Edition, (1998) John Wiley & Sons, Inc. Publication.
- Bioinformatics: Sequences, structure and databanks by Des Higgins and Willie Taylor, Oxford University Press, (2000).

BINF-MDC-1-101: INTRODUCTION TO BIOINFORMATICS – I (PRACTICAL)

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Searching literatures over PubMed.
2. Understanding and using Sequence information resources on web: EMBL-ENA,Genbank and DDBJ.
3. The information retrieval using Gquery.
4. Understanding and using Protein information resources on web: Uniprot
5. Performing BLASTp/n and interpreting its results.

Semester - II

BINF-MDC-2-201: INTRODUCTION TO BIOINFORMATICS - II

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

External Theory Marks: 40
Internal Theory Marks: 10
Total Theory Marks: 50
Exam Duration: 3 Hours

Course Duration:

1 Semester = 15 weeks which will have 30 hours of Theory Lectures (2 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 3 (2 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

The aim of this subject is to exposes the students to Bioinformatics sequence analysis and its applications.

General Instruction:

- The question paper will have seven questions, and each question have 08 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Multiple Sequence Alignment: SP (Sum of Pairs) measure to determine scoring function, Multiple Sequence Alignment methods: Progressive (Clustal Omega and PILEUP), Iterative and Block-based methods of MSA. Application of MSA.

Tools for MSA: Clustal Omega, Clustal-X and T-coffee.

UNIT - II

Phylogenetic analysis: Basic terminology in Phylogenetics, Introduction to phylogenetic analysis and its application, Phylogenetic tree terminology and topologies.

Methods of Phylogenetic Tree Construction:

Distance Based Methods: NJ, FM & UPGMA
Character Based Methods: Maximum Parsimony & Maximum Likelihood.
Phylogenetic Tree Evaluation Methods: Bootstrapping & Jackknifing.

Tools for Phylogenetic Analysis: Introduction to Phylip and MEGA packages.

Protein Structural databases:PDB, SCOP and CATH

Recommended Books:

- Introduction to Bioinformatics by V. Kothekar, 1st Edition, (2006), Dhruv Publications, Delhi.
- Bioinformatics: A modern approach by Vittal R. Srinivas, Prentice/ Hall of India Pvt. Ltd. New Delhi, (2005)
- Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
- Bioinformartdics: A practical guide to the analysis of genes and proteins. Ed. By
- Baxevanis, 3rd Edition, (1998) John Wiley & Sons, Inc. Publication.
- Bioinformatics: Sequences, structure and databanks by Des Higgins and Willie Taylor, Oxford University Press, (2000).

BINF-MDC-2-201: INTRODUCTION TO BIOINFORMATICS – II (PRACTICAL)

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

1. Performing multiple sequence alignment using Clustal Omega.
2. Building Phylogenetic tree using:
    a. Distance-based & character-based methods by PHYLIP or MEGA
    b. Tree evaluation
3. Understanding the PDB and its file format.
4. Downloading and visualizing protein structure from PDB.

Semester - III

BINF-MDC-3-301: BIOINFORMATICS METHODS FOR BIOMOLECULE ANALYSIS

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

External Theory Marks: 40
Internal Theory Marks: 10
Total Theory Marks: 50
Exam Duration: 3 Hours

Course Duration:

1 Semester = 15 weeks which will have 30 hours of Theory Lectures (2 hours/week)and 30 hours of practical classes (2 hours/week)
1 Credit = 15 hours of theory lectures or 30 hours of practical classes
Total Credits: 3 (2 Credits for Theory Lectures and 1 Credit for Practical)
Each credit is equal to 25 marks

Learning Outcomes:

Computational Methods for Gene and RNA Structure Analysisis to equip students with computational tools and methodologies for analyzing gene structure and predicting their structures and functions.

General Instruction:

- The question paper will have seven questions, and each question have 08 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT - I

Gene Prediction Methods: Searching by Signal, Searching by Content, HomologyBased Predictions, Markov Models, Hidden Markov Models in Gene Prediction.
Gene prediction tools: Genscan, Glimmer, Grail.

Promoter analysis: Concept of Promoter, predictions of promoter in prokaryoticand eukaryotic Genomes.

RNA Structure Prediction: Types of RNA secondary structure
Methods of RNA secondary structure prediction: Ab-initio and Homology based.

UNIT - II

Tools of RNA secondary structure prediction: Mfold and Vienna RNA package.

Protein Structure Determination Methods: Principle and Applications of X-raycrystallography and NMR.

Protein Structure Prediction:
Secondary Structure Prediction methods: CHAU FASMAN, GOR, Neural Network

Tertiary Structure Prediction methods: Homology Modeling, Threading/Foldrecognition and Ab-initio methods.

Recommended Books:

1. Jin Xiong (2006). Essential Bioinformatics, Cambridge University Press.
2. Bioinformatics: Sequences, structure and databanks by Des Higgins and Willie Taylor, Oxford University Press, (2000).
3. Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
4. Bioinformatics: A practical guide to the analysis of genes and proteins. Ed. By Baxevanis, 3.

BINF-MDC-3-301: BIOINFORMATICS METHODS FOR BIOMOLECULE ANALYSIS (PRACTICAL)

(Multi/Interdisciplinary Stream: Offered by Department of Bioinformatics for the students of other departments)

External Practical Marks:	20
Internal Practical Marks:	05
Total Practical Marks:	25

Gene Prediction Tools:

1. GenScan
2. GenMark

Protein Structure Modeling Using:

1. GOR and PSIPRED
2. Swiss Model
3. ProCheck

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Biotechnology

Semester-I

BIOT-MDC-1-101:-FOUNDATIONS OF BIOTECHNOLOGY-I

Learning Outcomes: -
This course will introduce the basic concepts of biotechnology to the students. They will learn about the history of biotechnology; the foundations of modern biotechnology; the role of biotechnology in fermentation industry, environment and modern medicine and the ethical implications of biotechnology. Students of other streams will develop basic skills related to biotechnology and will be able to communicate with persons skilled in the art.

External Theory	Internal Theory	External practical	Internal practical
45	05	22	03

Course Duration
30 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 2 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
Credits: 3 credits (Each credit is equal to 25 marks)

Instructions for the Paper Setters and Examiners:
Examiner will set a total of nine questions comprising two questions from each unit and one compulsory question of short answer type covering the whole syllabus. The students will attempt one question from each unit and the compulsory question. All questions may carry equal marks, unless specified.

Unit-I

Advent, scope and basics of biotechnology.
Study of model organisms: *E. coli*, *Saccharomyces cerevisiae*, *Drosophila melanogaster,*

Unit-II

Structure and function of the cell: the basic unit of life. Prokaryotic and Eukaryotic cells
Biomolecules in a cell (DNA, RNA and proteins) Enzymes -Classification, nomenclature and general properties

Unit-III

Green technology to controlpollution.
Introduction to basics of gene therapy.
Introduction to basic techniques like sterilization, centrifugation, electrophoresis, chromatography.

Unit-IV

Genetically modified organisms (GMOs) - transgenic plants and animals and their applications.
Public concerns and risks associated with GMOs
Ethical and social issues in biotechnology.

Reference Books:

1. Comprehensive Biotechnology, Murray Moo-Young, 2nd edition (2011), Pergamon Press.
2. Introduction to Biotechnology, William J. Thieman& Michael A. Palladino, 3rd edition (2012),Benjamin Cummings.
3. Biotechnology Expanding Horizons, BD Singh, 4th edition (2012), Kalyani Publishers.
4. The Ethics of Biotechnology (Biotechnology in the 21st Century), Jonathan Morris, Chelsea HousePub (L), 1st edition (2005).
5. Biotechnology, Applying the Genetic Revolution, David P. Clark & Nanette J. Pazdernik, 1st edition(2008), Academic Cell.
6. Molecular Biotechnology, Sandy B. Primrose, 2nd edition (1991), Blackwell Scientific Publications,ISBN.17
7. Biotechnology: Demystifying the concepts, David Bourgaize, Thomas R. Jewell & Rodolfo G. Buiser,1stedition(1999), Benjamin Cummings.
8. Ethical Issues in Biotechnology, Richard Sherlock & John D. Morrey, 1st edition (2002), Rowman&Littlefield Publishers.

BIOT-MDC-1-101: FOUNDATIONS OF BIOTECHNOLOGY-I (Practicals)

1. Concept of laminar flow hoods and Biosafety cabinets: principle and working.
2. Concept of Sterilization: Autoclave and Hot air oven (Principle and working).
3. Possible means of contamination and its control.
4. To perform membrane filtration.
5. To perform Paper Chromatography.
6. Introduction and use of equipment like electrophoretic apparatusand centrifuge.

Semester-I

BIOT-MDC-2-201: FOUNDATIONS OF BIOTECHNOLOGY-II

Learning Outcomes: This course will introduce the basic concepts of biotechnology to the students. Microbes play significant role in understanding Biotechnology industry so study of microbes frombasic to advance level, with understanding of biochemistry, cell structure and application makes this paper significant. The papers encompasses a wide range of disciplines aiming to better human life.

Course Duration

30 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 2 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
Credits: 3 credits (Each credit is equal to 25 marks)

External Theory	Internal Theory	External practical	Internal practical
45	05	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for the Paper Setters and Examiners:

Examiner will set a total of nine questions comprising two questions from each unit and one compulsory question of short answer type covering the whole syllabus. The students will attempt one question from each unit and the compulsory question. All questions may carry equal marks, unless specified.

UNIT-I

Carbohydrates: Classification and characteristics of Mono, Di, Polysaccharides. Epimers, Isomers, pyranose and furanose forms.
Reducing and non reducing sugars.

UNIT-II

Proteins: Classification of proteins according to their biological functions. Primary, secondary, tertiary and quaternary structure in proteins.
Enzymes: Classification and nomenclature and general properties of enzymes, steady state kinetics.

UNIT-III

Nucleic Acids: Structure of purine and pyrimidine bases. Nucleosides and nucleotides. Different forms of DNA and RNA. Physical and chemical properties of nucleic acids.

UNIT-IV

Lipids: Classification of lipids and fatty acids, general functions and structure of major lipid subclasses, acylglycerols, phosphoglycerols, phosphoglycerides, sphingolipids, glycosphingolipids. glycolipids.

BIOT-MDC-2-201: FOUNDATIONS OF BIOTECHNOLOGY–II (Practicals)

1. Preparation of physiological buffers.
2. Estimation of carbohydrates in given solution by Anthrone method.
3. Methods of protein estimation.
4. Methods for Lipid estimation.

Semester-III

BIOT-MDC-3-301: BIOANALYTICAL TECHNIQUES

Learning Outcomes: This paper will enable the students to learn important tools and techniques based on biophysical and biochemical principles so that they can understand application of these techniques in biotechnology and medical science. The paper will familiarize the students with the equipments found in the laboratories, their working and applications.

Course Duration
30 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 2 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)

Credits: 3 credits (Each credit is equal to 25 marks)

External Theory	Internal Theory	External practical	Internal practical
45	05	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for the Paper Setters and Examiners:
Examiner will set a total of nine questions comprising two questions from each unit and one compulsory question of short answer type covering the whole syllabus. The students will attempt one question from each unit and the compulsory question. All questions may carry equal marks, unless specified.

Unit – I

Principle, working and applications of Spectrophotometry (UV& visible)
Electrophoresis: Principle, types (Agarose gel, Native, SDS-PAGE) & their applications.

Unit – II

Microscopy: Properties; Light microscopy (Bright field Microscopy and Dark field Microscopy); Phase contrast and Fluorescent microscopy.
Electron microscopy: Scanning and Transmission Electron Microscopy.
Centrifugation: Principle, types (Differential, Density Gradient and Ultracentrifugation) & their applications.

Unit – III

Overview of chromatography: Principles and Applications of Ion exchange, Gel Permeation, Affinity Chromatography, HPLC.

Unit – IV

Radiotracers: GM counter, proportional and scintillation counters.
Mass spectrometry: Principle, Instrument used; ionization modes (Electro spray, Fast Atom Bombardment and Chemical ionization)

Reference Books:

1. K Wilson, J Walker (2000) Principles and techniques of biochemistry and Molecular Biology Seventh Edition, (Cambridge University Press).
2. SR Mikkelsen, E Cortin (2004). Bioanalytical Chemistry (2004) (John Wiley & Sons)
3. David Sheehan (2009). Physical Biochemistry: Principles and Applications, Second Edition. (John Wiley & Sons)
4. P. Carmona, R. Navarro, A, Hernanz (1997) Spectroscopy of biological molecules: Modern trends (Illustrated, Publisher Springer)
5. Bernard Valeur (2001). Molecular fluorescence: Principles and application. Wiley-VCH).Plenum USA.
6. Plummer D.T. (2004). An introduction to practical biochemistry (Tata McGraw Hill Publishers Co. Ltd., New Delhi).
7. Bansal, D.D., K Hardori, R & Gupta, M.M. (1985). Practical biochemistry (Standard Publication, Chandigarh).

BIOT-MDC-3-301: BIOANALYTICAL TECHNIQUES (Practicals)

1. Demonstration of SDS-PAGE.
2. Quantitative analysis of protein by UV/ Visible spectrophotometry by Bradford method.
3. Imparting knowledge, use and care of light microscope.
4. Gel filtration chromatography
5. Centrifugation of samples using Swing- bucket and Fixed angle rotor.

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Botany

Semester-I

Course Title: **PLANTS- IMPORTANCE AND PROPAGATION** **Course Code: BOT-MDC-1-101**

Learning Outcome: The course has been framed to familiarize students with the importance of plants for sustenance of our planet in general and mankind in particular. It also aims to educate them about different modes of plant reproduction.

Max. Marks: 75 **Course Credits =** 3
Theory: 75 Written Exam: 67 Internal Assessment: 8

Periods per week: Theory- Four of 45 minutes duration (Three Hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem solving, inductive and deductive methods and can use L.C.D., L.E.D., and multimedia projector in addition to black/white board.

UNIT – I 12 Hours

Diversity in habit (herb, shrub and tree) and habitat (terrestrial and aquatic), plant adaptations (hydrophytes and xerophytes), diversity on the basis of longevity (annual, biennials, perennials).

UNIT – II 11 Hours

Plants as source of: food (Wheat, Rice); timber (Shisham, Teak); medicine (Quinine, Sarpgandha); essential oils (Lemon Grass,Tulsi) and beverages (Tea and Coffee) - General morphology, botanical names and parts used.

UNIT – III 11 Hours

General account of asexual means of reproduction; Natural and artificial means of vegetative propagation; advantages and limitations; propagation by bulbs, corms, tubers, rhizomes, runners, stolons and suckers – general account; propagation by cutting, layering, grafting and budding – general account.

UNIT – IV 11 Hours

General account of sexual reproduction - structure of flower (calyx, corolla, androecium, Gynoecium) and its various forms (unisexual & bisexual), types of pollination (self vs. cross), fertilization; basic concept of dicot seed, structure and germination.

Suggested Readings:

1. Bhojwani, S.S., Bhatnagar, S.P. and Dantu, P.K. (2018). The embryology of angiosperms. (6th edition). Vikas Publishing House, Noida.
2. Coulter, J.M. (1903). Morphology of Angiosperms (Morphology of Spermatophytes. Part-II). Nebu press, pp 376.
3. Johri, B.M. and Srivastava, P.S. (2013). Reproductive Biology of Plants. Springer Science and Business Media, pp 320.
4. Khan, A.S. (2017). Flowering Plants: Structure and Industrial Products. Wiley; 1st edition, United Kingdom, pp 344.
5. Kochhar, S.L. (2016). Economic Botany. Cambridge University Press, pp 680.
6. Nanda, K.K. and Kochar V.K. (1985). Vegetative Propagation of Plants. Kalyani Publishers, New Delhi.
7. Ramawat, K.G. *et al*. (2014). Reproductive Biology of Plants. CRC Press, Boca Raton.
8. Sadhu, M.K. (1999). Plant Propagation. New Age International (P) Limited Publishers, New Delhi.

SEMESTER – II

Course Title: UTILIZATION AND CONSERVATION OF PLANTS **Course Code: BOT-MDC-2-201**

Learning Outcomes: The basic objective of this paper is to make the students aware of origin of plants, their civilization, utilization and extinction and conservation.

Max. Marks: 75 **Course Credits =** 3
Theory: 75 Written Exam: 67 Internal Assessment: 8

Periods per week: Theory – Four of 45 minutes duration (Three Hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Teaching Aids**:** Teachers can use lecture, lecture cum demonstration, project, assignments, problem solving, inductive and deductive methods, and can use L**.**C**.**D**.**, L**.**E**.**D**.**, Multimedia projector in addition to black**/**white board**.**

UNIT – I 12 Hours

Origin of agriculture-time and places of origin, domestication of cultivated plants; Forests resources –sustainable utilization and development; Ethnobotanyand its significance.

UNIT – II 11 Hours

General account, food and economic value of the plants as source of: Fibres (Cotton and Jute); Pulses (Mung, Chickpea); Spices and Condiments (Cumin, Fennel, Ginger and Turmeric); Vegetables and fruits (Potato, Tomato, Grapes and Mango).

UNIT – III 11 Hours

General account, parts used and uses of the plants as source of: Vegetable oils (Sunflower and Mustard); Pulp plants (bamboo); Medicinal plants (Ashwagandha, Aloe vera); brief account of resins and gums; psychoactive drugs (Hemp and Poppy).

UNIT – IV 11 Hours

IUCN categories of plants; plant extinction – causes and preventive measures; principles of conservation; in situ and ex situ conservation strategies; Conventions on Biodiversity (CBD); International agreement to protect species and habitat; Brief account of National Parks, Biosphere Reserves and Sanctuaries.

Suggested Readings:

1. Akeroyd, J. and Jackson, P.W. (1995). A Handbook of Botanic Garden and Reintroduction of Plants to the Wild. Botanic garden conservation Union, UK.
2. Chowdhery, H.J. and Murty, S.K. (2000). Plant Diversity and Conservation in India – an overview. Bishen Singh Mahendra Pal Singh, Dehradun.
3. Directory of Indian Wetlands (1993). W.W.F. India, New Delhi and AWB, Kuala Lumpur.
4. Heywood, V. (1995). Global Biodiversity Assessment. United National Environment Programme. Cambridge University Press, Cambridge, U.K.
5. Haunter, M.L. and Gibbs, J. (2007). Fundamentals of Conservation Biology. 3rd Edn. Blackwell Publishing, U.K.
6. Kothari, A. (1997). Understanding Biodiversity: Life Sustainability and Equity. Orient Longman.
7. Meffe, G.K. and Ronald, C.R. (1994). Principles of Conservation Biology. Sinauer Associates. INC Publishers, USA.
8. Primack, R.E. (2006). Essentials of Conservation Biology. 4th Edn. Sinauer Associates, U.S.A.
9. Rodgers, N.A. and Panwar, H.S. (1988). Planning a Wildlife Protected Area Network in India. Vol. I. The Report Wildlife Institute of India, Dehradun.

SEMESTER – III

Course Title: NATURAL RESOURCE MANAGEMENT **Course Code:BOT-MDC-3-301**

Learning Outcome: The objective of this course is to provide knowledge to the students on importance, sustainable utilization, national and international efforts for conservation and management of natural resources.

Max. Marks: 75 **Course Credits =** 3
Theory: 75 Written Exam: 67 Internal Assessment: 8

Periods per week: Theory- Four of 45 minutes duration (Three hours)

Instructions for Paper Setters:

Question Paper will have four units. Students will attempt **five** questions out of total nine comprising of one **COMPULSORY** question consisting of 15 questions having 8 MCQ and 7 fill in the blanks of one mark each covering the whole syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions of 13 marks each and one question is to be attempted out of each unit.

Instruction Aids: Teaching methodology includes series of lectures making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc.

UNIT – I 12 Hours

Natural Resources: Definition and types and conservation of natural resources, conventional (Coal, petroleum, natural gas and electricity) and non-conventional (solar energy, tidal energy, wind energy, geothermal energy) energy sources, alternate emerging energy resources for future (Petroplants, biodiesel).

UNIT – II 11 Hours

Water Resources: Types of water sources, Water resource scenario in India, Sustainable water use and Water Conservation, rainwater harvesting and watershed management.

UNIT – III 11 Hours

Forests Resources: Significance of forests, Major forest types of India, Impact of Mining on Forests, Dams and their impact on forests, Deforestation, Conservation of forest resources, Forest Policy of India, Joint Forest management (JFM).

UNIT – IV 11 Hours

Soil Resource: Land use in India, Soils of India, Soil Degradation and its control, Water logging and salinity, Soil erosion, Wasteland Reclamation. Contemporary practices in resource management.

Suggested Readings:

1. Sharma, P.D. (2007). Ecology and Environment. Rastogi Publications, Meerut.
2. Vasudevan, N. (2006). Essentials of Environmental Science. Narosa Publishing House, New Delhi.
3. Singh, J. S., Singh, S.P. and Gupta, S. (2006). Ecology, Environment and Resource Conservation. Anamaya Publications, New Delhi.
4. Pandey S.N. and Misra, S.P. (2016) Environment and Ecology. Ane Books Pvt. Ltd. New Delhi.
5. Rogers, P.P., Jalal, K.F. and Boyd, J.A. (2008). An Introduction to Sustainable Development. Prentice Hall of India Private Limited, New Delhi.

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Chemistry

Semester	**I**
Code of Paper	**CHM-IDC-1**
Title of Paper	**Basic Analytical Methods-1**
Total Credit	**03 (02 Theory + 01 Practical)**
Total Marks	**75 (50 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 45** **Internal Marks = 05**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 2 Hrs. per week** **Practical: 2 Hrs. per week**
Applicable to all students who have opted for Table 1 and Table 2	
Note: - This is a three credit introductory course. It is not the same as the courses studied by the students at the 12th level. It does not belong to the core Chemistry. It is an interdisciplinary and can be useful for all the disciplines of Science, Humanities, Commerce, etc. The syllabus can be taught by teachers from any discipline who knows the subject. Therefore, these courses may be offered to students of all disciplines irrespective of the subjects studied at the 12th level.	
LEARNING OUTCOMES SPECIFIC TO DISCIPLINE The course will provide following outcomes. 1. Development of a comprehensive understanding of sampling, error analysis, and both volumetric and gravimetric analytical techniques. 2. Students will be equipped with practical skills for collecting samples, evaluating data, and performing laboratory analyses with precision and accuracy.	
GENERIC OUTCOMES 1. **The students will be able to learn Critical Thinking and Problem Solving, Practical Laboratory Skills, Data Management and Evaluation, Safety and Hazard Management, Communication Skills,** Ethical and Professional Responsibility, **Interdisciplinary Application of Knowledge** 2. These generic outcomes equip students with a broad set of transferable skills, making them capable of conducting rigorous scientific analysis while working safely, ethically, and effectively in collaborative environments.	

Semester-1
Syllabus for Theory Paper
CHM-IDC-1
Basic Analytical Methods-1

End semester = 45	**Credits - 02**
Internal Marks = 05	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

Unit I: Sample collection and data Evaluation (15hours)

Sampling, Collection and Preservation of Samples (6 Hrs.)
Sampling and theories of sampling, Sampling procedure, Collectionof Samples and its storage, Sampling of gases, liquids and solids, Preparation of samples for analysis, calibration standards, Hazards in sampling, simple equilibrium calculations, General methods of preservation.

Errors and Statistics (9 Hrs.)
Classification of errors, Minimization of errors, Accuracy, Precision, Distribution of random errors, errors in addition and multiplication, significant figures, Mean and standard deviation, rejection of measurements, variance, numerical problems related to evaluation of analytical data.

Unit II: Lab methodsofanalysis(15 hours)

Volumetric methods (8 Hrs.)
Theory of volumetric methods of analysis, preparation of solutions, equivalence point, standard solutions, Primary and Secondary standards, end point detection, theory of indicators and their selection for volumetric analysis, limitations, interferences, types of titrations, their theory and applications - acid base titrations, redox titrations, complexometric titrations (EDTA titrations).

Gravimetric methods (7 Hrs.)
Introduction and theory, precipitation methods, Tyndall effect, colloid formation, supersaturation and precipitation, Co-precipitation, conditions of precipitation, precipitation as hydroxides, oxalates, phosphates and sulphates, washing of precipitates, applications of gravimetry in separation, organic precipitants, DMG, 8-hydroxyquinoline.

Books/Literature suggested

1. Vogel's Textbook of quantitative chemical analysis, 5th Ed., Jeffery, G.H., Bassett, J., Mendham, J., Denney, R.C.
2. Principles of Inorganic Chemistry, Puri Sharma Kalia, Vishal Publishing Co.
3. University Practical Chemistry, Kamboj, P.C. 2nd edition, Vishal Publishing Co.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1** and **UNIT-2** and a **compulsory Unit**.
2. The **UNIT-1**and**UNIT-2** will have two questions each of **15 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **15 marks**) will consist of two types of questions:**5 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **three questions in all:** the compulsory question and one question each from the UNIT-1 and UNIT-2.
2. The use of scientific calculators is not allowed.

Semester-1
Syllabus for Practical Paper
CHM-IDC-1
Basic Analytical Methods-1

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. Volumetric analysis

1. Calibration of apparatus and calculation of %age error
2. Preparation of standard solutions.
3. Acid-base titrations
    a. HClvsNaOH
    b. Na_2CO_3vsHCl
    c. Mixture of bases versus HCl

II. Gravimetric analysis

4. Analysis of following metal ions
    a. Ni^{2+} as Ni-DMG complex
    b. Cu^{2+} as copper(I)-thiocyanate
    c. Ba^{2+} as barium(II) sulphate

Books/Literature suggested

1. Vogel's Textbook of quantitative chemical analysis, 5th Ed., Jeffery, G.H., Bassett, J., Mendham, J., Denney, R.C.
2. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition, Pubs: R.Chand& Co., New Delhi, 2002.

Semester	**II**
Code of Paper	**CHM-IDC-2**
Title of Paper	**Basic Analytical Methods-2**
Total Credit	**03 (02 Theory + 01 Practical)**
Total Marks	**75 (50 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 45** **Internal Marks = 05**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 2 Hrs. per week** **Practical: 2 Hrs. per week**
Applicable to all the students who have opted Table 1 or Table 2	

Note: - This is a three credit introductory course. It is not the same as the courses studied by the students at the 12th level. It does not belong to the core Chemistry. It is an interdisciplinary and can be useful for all the disciplines of Science, Humanities, Commerce, etc. The syllabus can be taught by teachers from any discipline who knows the subject. Therefore, these courses may be offered to students of all disciplines irrespective of the subjects studied at the 12th level.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

1. Students will develop the Understanding of the history and evolution of chromatography as an analytical technique.
2. Students will be able to ddifferentiate between analytical and preparative chromatography and their respective applications.
3. Students will be able to define key terms used in chromatography, including mobile phase, stationary phase, eluent, analyte, elute, and distribution coefficient.
4. Students will learn the classification of chromatography techniques based on principles of separation, shape of stationary phase, and polarity of phases.
5. Students will understand the principle behind thin layer chromatography (TLC) and mechanism of column chromatography, including the separation of components based on their differential interactions with the stationary and mobile phases.

GENERIC OUTCOMES

1. Students will learn Analytical and Practical Skills,**Understanding of Principles, Problem Solving and Innovation, Data Analysis and Evaluation, Communication and Reporting, Safety and Environmental Awareness, Interdisciplinary Application**
2. These outcomes will ensure that students develop a comprehensive understanding of chromatography and can apply these techniques in both theoretical and practical scenarios, equipping them with skills that are essential for modern scientific work.

Semester-2
Syllabus for Theory Paper
CHM-IDC-2
Basic Analytical Methods-2

End semester = 45	**Credits - 02**
Internal Marks = 05	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

Unit I: Basic concepts of Chromatography(15hours)

Introduction to chromatography (7Hrs)
Description and background, analytical and preparative chromatography, important terms used in chromatography – mobile phase, stationary phase, eluent, analyte, elute, distribution coefficient, classification of chromatography based on principles of separation, shape of stationary phase, polarity of phases, advantages and applications of chromatography.

Paper chromatography(8Hrs)
Principle, papers as a chromatographic medium, modified papers, solvent systems, mechanism of paper chromatography, experimental technique, different development methods-ascending, descending, horizontal, circular spreading, multiple development, visualization and evaluation of chromatograms, Rf values, applications.

Unit II: Column and thin layer chromatography(15 hours)

Thin layer chromatography (8 Hrs.)
Principle, stationary phase materials, applications, activation of adsorbent, sample development, solvent systems, development of chromatoplate, visualization methods and applications in the separation.

Column chromatography (7 Hrs.)
Principle,columns,matrixmaterials,stationaryphase,columnpacking,applicationofsample,columndevelopmentandsample elution,detectorsandfractioncollectors, applications, Ion exchange column chromatography, Ion exchange resins. Applications of ion exchange chromatography.

Books/Literature suggested

1. Vogel's Textbook of quantitative chemical analysis, 5th Ed., Jeffery, G.H., Bassett, J., Mendham, J., Denney, R.C.
2. Vogel, A.I., Tatchell, A.R., Furnis, B.S., Hannaford, A.J., & Smith, P.W.G., *Text book of Practical Organic Chemistry*,Prentice-Hall, 5th edition, 1996.
3. Mendham, J., *A.I. Vogel's Quantitative Chemical Analysis***,** 6th Ed., Pearson, 2009.

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1** and **UNIT-2** and a **compulsory Unit**.
2. The **UNIT-1**and**UNIT-2** will have two questions each of **15 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit**(of **15 marks**) will consist of two types of questions:**5 multiple-choice questions**of one mark each and**5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **three questions in all:** the compulsory question and one question each from the UNIT-1 and UNIT-2.
2. The use of scientific calculators is not allowed.

Semester-2
Syllabus for Practical Paper
CHM-IDC-2
Basic Analytical Methods-2

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. Paper chromatography

1. Separation of dyes
2. Separation of carbohydrate molecules

II. Thin layer chromatography

3. Preparation of TLC plates
4. Separation and identification of Ni^{2+} and Co^{3+} ions
5. Separation of inks
6. Determination of Rf values and identification of organic compounds
    a. Separation of isomeric mixture of ortho and para-nitroaniline using hexane and ethyl acetate (8.5:1.5) as the solvent system by thin layer chromatography.
    b. Separation of benzaldehyde and nitrobenzene using $KMnO_4$ solution as a visualizing agent.

III. Column chromatography

7. Separation of leaf pigments

Books/Literature suggested

1. Vogel's Textbook of quantitative chemical analysis, 5th Ed., Jeffery, G.H., Bassett, J., Mendham, J., Denney, R.C.
2. Khosla, B.D., Garg, V.C., Gulati, A., Senior Practical Physical Chemistry; 11th edition.
3. Vogel, A.I., Tatchell, A.R., Furnis, B.S., Hannaford, A.J., & Smith, P.W.G., *Textbook of Practical Organic Chemistry*, Prentice-Hall, 5th edition, 1996.
4. Mendham, J., *A.I.Vogel's Quantitative Chemical Analysis*, 6thEd., Pearson, 2009.

Semester	**III**
Code of Paper	**CHM-IDC-3**
Title of Paper	**Colours and dyes**
Total Credit	**03 (02 Theory + 01 Practical)**
Total Marks	**75 (50 Theory + 25 Practical)**
Marks distribution of the Theory paper	**End semester = 45** **Internal Marks = 05**
Marks distribution of the Practical paper	**End semester = 22** **Internal Marks = 03**
Contact Hours	**Theory: 2 Hrs. per week** **Practical: 2 Hrs. per week**

Applicable to all the students who have opted for Table 1 or Table 2

Note: - This is a three credit introductory course. It is not the same as the courses studied by the students at the 12th level. It does not belong to the core Chemistry. It is an interdisciplinary and can be useful for all the disciplines of Science, Humanities, Commerce, etc. The syllabus can be taught by teachers from any discipline who knows the subject. Therefore, these courses may be offered to students of all disciplines irrespective of the subjects studied at the 12th level.

LEARNING OUTCOMES SPECIFIC TO DISCIPLINE

1. Students will be able to understand the basic concept of colour and its perception by the human eye.
2. Students will explore the early history of pigments and dyes, including their cultural significance and historical applications. Learn the process of dyeing and its historical evolution, including traditional and modern methods.
3. Students will be identifying the different types of natural dyes and their applications in various industries.
4. Students will gain hands-on knowledge of the synthesis of key synthetic dyes, including Methyl Orange, Prussian Blue, Chrome Yellow, Chrome Green, Organol Brown, and Aniline Yellow, and their chemical reactions involved.
5. Students will understand the sources and extraction processes of natural dyes from plant, animal, and mineral sources.· Understand the basic concept of pigments, including their role in producing colour in both natural and synthetic forms.

GENERIC OUTCOMES

1. **Students will gain Fundamental Knowledge of Colour Science, Dye and Pigment Synthesis and Application, Environmental and Sustainability Awareness,** Practical Laboratory Skills.

Semester-3
Syllabus for Theory Paper
CHM-IDC-3
Colours and Dyes

End semester = 45	**Credits - 02**
Internal Marks = 05	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

Unit I: Introduction to colours and dyes (15 hrs.)

Introduction to Colours (7 Hrs.)
Introduction to color, The Early History of Pigments and Dyes, Recent and Current Trends in Colour Chemistry, Introduction to Physical and Chemical Basis of Colour, Visible Light, The Eye, The Causes of Colour, The Interaction of Light with Objects.

Classification and Synthesis of Dyes (8 Hrs.)
Dyeing, History of dyeing, Types of natural dyes, Synthetic dyes, Properties of synthetic dyes, Classification of synthetic dyes, Synthesis of Methyl Orange, Prussian blue, Chrome yellow, Chrome green, Organol brown, Aniline yellow.

Unit II: Dyes and Pigments (15 hrs.)

Dyes and Green Alternatives (8 Hrs.)
Sources and Extraction of Natural Dyes, Textile Dyes, Environment and Sustainability concerns, Natural Dyeing Processes.

Pigments(7 Hrs.)
Introduction to pigments, inorganic pigments, organic pigments, Pigments for Special Effects, Colour in decorative cosmetics, high technology dyes and pigments, applications in real life.

Books/Literature suggested

1. Color Chemistry, Christie, R.M. (RSC publisher).
2. Fundamental Processes of Dye Chemistry by Hans Edward Fierz-David & Louis Blangey (Interscience Publishers).
3. A text book of dye chemistry (The Chemistry of dye stuffs) by Dr. GeorgvonGeorgievics& Dr. Eug. Grandmougi (Unwin Bros. Ltd.).
4. Color for Art, Science and Technology by Azimuth, edited by Kurt Nassau (Elsevier Science B. V.).

Instructions for the examiner

1. The examiner will set the question paper comprising **UNIT-1** and **UNIT-2** and a **compulsory Unit**.
2. The **UNIT-1** and **UNIT-2** will have two questions each of **15 marks**. Each question shall have 3-4 questions as sub-parts equally distributed within the syllabus of the respective unit.
3. The **compulsory unit** (of **15 marks**) will consist of two types of questions: **5 multiple-choice questions** of one mark each and **5 short-answer** type questions of two marks each.

Instructions for students

1. The students will attempt **three questions in all:** the compulsory question and one question each from the UNIT-1 and UNIT-2.
2. The use of scientific calculators is not allowed.

Semester-3
Syllabus for Practical Paper
CHM-IDC-3
Colours and Dyes

End semester = 22	**Credit - 01**
Internal Marks = 03	**Contact Hours – 2/week**
Examination Time allowed	**3 Hours**

I. Preparation of following

a. Methyl Orange
b. Prussian blue
c. Chrome yellow
d. Chrome green
e. Organol brown
f. Aniline yellow

II. Hands on practice on the following.

a. Dyeing of fabric with synthetic and natural dyes
b. Study of effect of pH effect on colour of indicators/ dyes
c. Dye degradation using activated charcoal

Books/Literature suggested

1. Handbook on Natural Dyes for Industrial Applications (Extraction of Dyestuff from Flowers, Leaves, Vegetables) 2nd Revised Edition by Dr. Padma S Vankar.
2. b. Traditional methods described in Natural Dyes and their application to textiles by
3. Dr. Padma S Vankar.Green Chemistry Experiments by Royal Society of Chemistry.
4. Laboratory Manual for Principles of General Chemistry by Slowinski et al.
5. Fundamentals of Analytical Chemistry by Skoog, West, Holler and Crouch.
6. Dye Chemistry-Exploring Colour from nature to lab-Intech Open.
7. Fundamental Processes of Dye Chemistry by Hans Edward Fierz-David & Louis Blangey (Interscience Publishers).
8. Vogel's Text Book of Practical Organic Chemistry, 5th Edition, Brian S. Furniss Antony J. Hannaford Peter W. G. Smith Austin R. Tatchell.
9. Vogel's Textbook of Quantitative Chemical Analysis 6th Edition, J Mendham, R.C. Denney, J.D. Barnes, M. Thomas, B. Sivasankar

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Fashion Designing

Semester-I

INTRODUCTION TO FAMILY AND COMMUNITY SCIENCE
BSC-FD-MDC-1-101

THEORY

Credit: 2
Duration of Class: 2 hours per week
Duration of Exam: 3 hours
Assessment: 10

Maximum Marks: 50
Paper: 40
Internal

Instructions to the Examiner

- Question paper will have four sections/units. Paper setter will set a total of nine questions comprising of two questions from each unit and one compulsory question of short answer type covering the whole syllabus.
- Student will attempt a total of five questions in all, selecting one question from each section/units including the compulsory question.
- All questions may carry equal marks, unless specified.

Learning Outcome:

Students will have an in depth understanding of the following:

- Understanding of the self and one's role and responsibilities as a productive individual and as a member of family, community and society.
- Discipline of Family and Community Science for professional careers.
- Knowledge specific to five domains namely, Foods and nutrition, Human Development, Clothing and Textiles, Family Resource Management and Extension Education.
- Functional skills in the five domains for career and employment

UNIT I

1. Introduction & Definition of Family and Community Science
   - Core areas of Family and Community Science
   - Scope & opportunity in Family and Community Science
2. Development of Family and Community Science as a discipline

UNIT II

3. Foods & Nutrition
   - Basic terminology- Food, Nutrition, Food groups, Balance Diet, Food preservation, Meal planning, Nutritional deficiencies and Food hygiene
   - Role & Importance of Foods & Nutrition in Community
   - Functions of food
   - Basic Food groups: sources and their functions
4. Human Development
   - Basic terminology- Growth, Development, Childhood, Adolescent, Anxiety, Expression, Fear, Personality, Sex Education
   - Role & Importance of Human Development in Community
   - Stages of growth and development

UNIT III

5. Clothing & Textiles
    - Basic terminology- Fiber, Yarn, Spinning, Fabric, Apparel, Textiles, Laundry, Starch, Blues, Weaving, Knitting, Non-wovens
    - Role & Importance of Clothing & Textiles in Community
    - Care and maintenance of garments.
6. Resource Management
    - Basic terminology- Home Management, Human resources, Time management, Work simplification, Ergonomics, Work ethics, Family Budget, Furniture & Furnishings, Floor coverings
    - Role & Importance of Resource Management in Community
    - Time, energy and money management

UNIT IV

7. Extension & Communication
    - Concept, Nature & Scope of Extension
    - Family and Community Science Extension as a Discipline and its contribution towards development
    - Methods & Media of Community Outreach
    - Role & Importance of Extension & Communication in Community and National Development.

Recommended Readings

1. Basic Food Preparation: A Complete Manual, Faculty of Lady Irwin College, Orient Longman publishers.
2. Bhatnagar. O.P &Dhama, O.P. (2009). Education And Communication For Development 2ed. New Delhi: Oxford & IBH Publishing Co. Pvt Ltd. ISBN-13: 978-8120400306.
3. Dantyagi S, “Fundamentals of Textiles and their Care,” Orient Longman Ltd, New Delhi.
4. Goyal, M. (2022). Home Science. SBPD Publishers
5. R Vatsala "Textbook of Textiles & Clothing" (2003), ICAR, New Delhi
6. Seetharaman P, Batra S. and Mehra P 2005. An Introduction to Family Resource Management. CBS Publishers and distributors, New Delhi.
7. Sumati R Mudambi& MV Rajagopal (2004). Fundamentals of Foods & Nutrition. Forth edition, New Age P Publishers.

Semester-I

INTRODUCTION TO FAMILY AND COMMUNITY SCIENCE
BSC-FD-MDC-1-101
PRACTICAL

Credit: 1
Duration of class: 2 hours per week
Duration of Exam: 3 hours
Assessment: 05

Maximum Marks: 25
Paper: 20
Internal

Instructions to examiner

- The external and internal examiners are requested to be present at the examination centre before the commencement of the examination.

Learning Outcome:

Students will have an in depth understanding of the following:

- Develop an understanding of the self and one's role and responsibilities as a productive individual and as a member of family, community and society.
- Learning across diverse domains and undertake a critical analysis of issues and concerns specific to family, community and society.
- Learners with the basic knowledge specific to five domains namely, Foods and nutrition, Human Development and Family studies, Fabric and Apparel, Resource Management and Communication and Extension.

Content

1. Preparation of Audio–Visual Aids on following topics-
    - Food Pyramid
    - Menu Planning for different occasions
    - Teaching aids for pre-school children
    - Table setting
    - Textile labels
2. Planning & Arrangement of various thematic parties (Decoration, Table setting, Meal planning and Fashion styling)

Note- Marks will be awarded by External on the basis of sessional work presentation

Recommended R e a d i n g s

1. Sumati R Mudambi & MV Rajagopal (2004). Fundamentals of Foods & Nutrition. Forth edition, New Age P Publishers.
2. Basic Food Preparation: A Complete Manual, Faculty of Lady Irwin College, Orient Longman publishers, 2008.
3. Seetharaman P, Batra S. and Mehra P 2005. An Introduction to Family Resource Management. CBS Publishers and distributors, New Delhi.
4. Bhatnagar. O.P &Dhama, O.P. (2009). Education And Communication For Development 2ed. New Delhi: Oxford & IBH Publishing Co. Pvt Ltd. ISBN-13: 978-8120400306.
5. Goyal, M. (2022). Home Science. SBPD Publishers
6. Dantyagi S, "Fundamentals of Textiles and their Care," Orient Longman Ltd, New Delhi.
7. R Vatsala "Textbook of Textiles & Clothing" (2003), ICAR, New Delhi

Semester-II

SUSTAINABILITY IN EVERYDAY LIFE
BSC-FD-MDC-2-201
THEORY

Credit: 2 **Maximum Marks: 50**
Duration of Class: 2 hours per week **Paper: 40**
Duration of Exam: 3hours **Internal Assessment: 10**

Instructions to the Examiner

- Question paper will have four sections. Paper setter will set a total of nine questions comprising of one question from each unit and one compulsory questionof short answer type covering the whole syllabus.
- Student will attempt a total of five questions in all, selecting one question from each section/units including the compulsory question.
- All questions may carry equal marks, unless specified.

Learning Outcome:
Students will have an in depth understanding of the following:

- Terms related to sustainability
- Sustainable practices at home

Content:

UNIT –I

1. Terminology: Sustainability, 3R's of sustainability, Eco-labels, Biodegradable and Non Biodegradable waste, E-Waste , Zero waste concept, Carbon Foot Print, Fair trade, Green washing, Life cycle assessment, Circular economy

UNIT –II

2. Sustainable Development: Definition and Concept
3. Five P's of Sustainable Development
   People, planet, prosperity, peace and partnership
4. Factors of Sustainability

UNIT –III

5. Sustainable practices at home
6. Different ways to be sustainable in everyday life
7. Sustainable brands in everyday life

UNIT –IV

8. Waste management: Definition and Types
9. Concept of Waste management
10. Government Initiatives for Sustainable development

Recommended readings:

1. Cradle to Cradle: Remaking the Way We Make Things By William McDonough & Michael Braungart
2. The Upcycle: Beyond Sustainability – Designing for Abundance by William McDonough & Michael Braungart
3. Grow Food For Free: The sustainable, zero-cost, low-effort way to a bountiful harvest by Huw Richards
4. Strategy for Sustainability: A Business Manifesto by Adam Werbach
5. Zero Waste Sewing: 16 projects to make, wear and enjoy by Elizabeth M Haywood Cooatalaa Press
6. 50 Nifty Thrifty Upcycled Fashions: Sew Something from Nothing by Cynthia Anderson, Stackpole Books
7. Wear, Repair, Repurpose: A Maker's Guide to Mending and Upcycling Clothes by Lily Fulop, Countryman Press
8. Recycling in Textiles by YoujiangWang , Woodhead Publishing Ltd
9. Wise Craft Quilts: A Guide to Turning Beloved Fabrics Into Meaningful Patchwork by Blair Stocker, Roost Books
10. Creative Recycling in Embroidery by Val Holmes **,** Batsford Ltd
11. Waste Recycling and Reuse by Sterling **,** Sterling Publishers
12. Reduce, Reuse, Recycle: An Easy Household Guide (Green Books Guides) by Nicky Scott (Author), Axel Scheffler **,** Green Books
13. The Complete Technology Book on E-Waste Recycling ,NPCS Board of Consultants & Engineers, Asia Pacific Business Press Inc.
14. Household Waste Recycling by Dr. Richard Waite **,** Earthscan Ltd

Semester-II

SUSTAINABILITY IN EVERYDAY LIFE
BSC-FD-MDC-2-201
PRACTICAL

Credit: 1 **Maximum Marks: 25**
Duration of Class: 2 hours per week **Paper: 20**
Duration of Exam: 3hours **Internal Assessment: 05**

Instructions to examiner:

- The external and internal examiners are requested to be present at the examination centre before the commencement of the examination.

Learning Outcome

Students will have an in depth understanding of the following:

- Understand Terms related to sustainability
- Sustainable practices at home

Content:

1. Preparation of Audio- visual aids on sustainability and sustainable practices.
2. Awareness generation on sustainability and sustainability practices through Prepared audio-visual aids in neighbourhood (On Campus and off Campus).
3. Product development from waste (5 different products will be developed from different mediums).
4. Preparation of file and presentation on field work.

Recommended readings:

1. Cradle to Cradle: Remaking the Way We Make Things By William McDonough & Michael Braungart
2. The Upcycle: Beyond Sustainability – Designing for Abundance by William McDonough & Michael Braungart
3. Grow Food For Free: The sustainable, zero-cost, low-effort way to a bountiful harvest by Huw Richards
4. Strategy for Sustainability: A Business Manifesto by Adam Werbach
5. Zero Waste Sewing: 16 projects to make, wear and enjoy by Elizabeth M Haywood Cooatalaa Press
6. 50 Nifty Thrifty Upcycled Fashions: Sew Something from Nothing by Cynthia Anderson, Stackpole Books
7. Wear, Repair, Repurpose: A Maker's Guide to Mending and Upcycling Clothes by Lily Fulop, Countryman Press
8. Recycling in Textiles by YoujiangWang , Woodhead Publishing Ltd
9. Wise Craft Quilts: A Guide to Turning Beloved Fabrics Into Meaningful Patchwork by Blair Stocker, Roost Books
10. Creative Recycling in Embroidery by Val Holmes , Batsford Ltd
11. Waste Recycling and Reuse by Sterling , Sterling Publishers
12. Reduce, Reuse, Recycle: An Easy Household Guide (Green Books Guides) by Nicky Scott (Author), Axel Scheffler , Green Books
13. The Complete Technology Book on E-Waste Recycling ,NPCS Board of Consultants & Engineers, Asia Pacific Business Press Inc.
14. Household Waste Recycling by Dr. Richard Waite , Earthscan Ltd

Semester-III

FOLK ARTS OF INDIA (MULTI DISCIPLINARY COURSE)
BSC-FD-MDC-3-301
THEORY

Credit: 2 **Maximum Marks: 50**
Duration of Class: 2hrs/week **Paper: 40**
Duration of Exam: 3hrs **Internal Assessment: 10**

Instructions to Examiner:

- Question paper will have four sections/units. Paper setter will set a total of nine questions comprising of two questions from each unit and one compulsory question of short answer type covering the whole syllabus.
- Student will attempt a total of five questions in all, selecting one question from each section/units including the compulsory question.
- All questions may carry equal marks, unless specified.

Learning Outcome:

Students will have an in depth understanding of the following:

- Various Folk arts of India
- Process and significance of various folk arts

Content:

Note: Study the following with special reference to origin, motif & design, colour & technique and significance of various folk arts and their diversification.

Unit-I

1. Pichwai Paintings - Rajasthan
2. Sanjhi Art - Uttar Pradesh
3. Rogan work of Kutch Gujrat
4. Pahari Miniature paintings of Himachal Pradesh

Unit-II

5. Bhil Paintings - Madhya Pradesh
6. Gond Paintings - Madhya Pradesh and Chhattisgarh
7. Mandana Paintings - Rajasthan and Madhya Pradesh
8. Warli Paintings - Maharashtra

Unit-III

9. Thangka Paintings – Sikkim
10. Madhubani Paintings – Bihar
11. Sohrai Paintings - Jharkhand and Bihar
12. Aripana – Bihar

Unit-IV

13. Alpana - West Bengal, Tripura, Assam
14. Pattachitra of Orissa
15. Kaali ghat paintings of West Bengal
16. Santhal art of West Bengal

Recommended Readings:

1. Randhawa. M. S. “Indian Miniature Paintings”, Roli Books International, New Delhi 1981
2. Shrikant. Usha , “Designs for Lifetime”, PriyavratGoenka, The South Pacific Clothes Export Company Pvt. Ltd. Andheri West , Mumbai 2002
3. Dayal. Bharti, “Indian Art Series: Madhubani Art”, Niyogi Books, Noyogi Offset Pvt Ltd. New Delhi 2015
4. Lehri. M. S., “Indi-Folk: Folk and Tribal Designs of India” , Super Book House, Mumbai India, 2009
5. Satyawadi. S. “Unique Art of WarliPaintings”,D.K.Printworld (P) Ltd. ,New Delhi 2010
6. Taubes. F. “A Guide to Traditional and Modern Painting”, Murray Printing Co., Great Britania 1963
7. Goswamy B.N. “Pahari Painting of the Nala - Damyanti”, National Museum, New Delhi 1975
8. Randhawa. M.S. “Kangra Valley Painitng”, Ministry of Information and Broadcasting, Govt. Of India, Times of India, Bombay
9. Ebeling. K. “RagmalaPainitng”, Ravi Kumar, Basel, Paris, New Delhi 1973
10. Gillow. J. & Barnard N. “Indian Textiles”, Om Books International, New Delhi 2008
11. Hataraba. K. “Textile Parts of India” Cronicle Books, Kyoto Shoin Co. Ltd., Japan 1996
12. Aryan. K.C. “Basis of Decorative Elements in Indian art”, Rekha Prakashan, New Delhi 1981
13. Chishti. R.K.&Jain.R,“Hand Crafted Indian Textiles” Roli Books Pvt. Ltd.,Lusture Press, New Delhi, India, 2000
14. Mode.H. & Chandra. S., “Indian Folk Art”, Alpine Fine Arts Collection Ltd., New York 1985
15. “An Introduction to Indian Art”, Textbook in Fine Arts for Class XI, NCERT, New Delhi 2012
16. “Living Traditional Tribal & Folk painting of India”, Centre for Cultural Resources & Training, Ministry of Culture, Govt. of India, New Delhi
17. “Living Craft Tradition of India”, NCERT, New Delhi

Semester-III

FOLK ARTS OF INDIA
BSC-FD-MDC-3-301
PRACTICAL

Credit: 1 **Maximum Marks: 25**
Duration of Class: 2hrs/week **Paper: 20**
Duration of Exam: 3hrs **Internal Assessment: 05**

Instructions to examiner:

- The external and internal examiners are requested to be present at the examination centre before the commencement of the examination.

Learning Outcome:

To enable students to understand:

- The technique of various Folk arts of India
- Various materials, motifs and colour combinations used in different folk arts

Content:

I. Collection of motifs of various folk arts and learning the technique and making samples of the following arts:
   a) Pichwai Paintings - Rajasthan
   b) Rogan Art of Gujarat
   c) Bhil Paintings - Madhya Pradesh
   d) Gond Paintings - Madhya Pradesh and Chhattisgarh
   e) Mandana Paintings - Rajasthan and Madhya Pradesh
   f) Warli Paintings - Maharashtra
   g) Madhubani Paintings – Bihar
   h) Pattachitra of Orissa

II. Making article from any one of the above arts
(wall panels, stole, dupatta, tote bag, cushion covers)

Recommended Readings:

1. Randhawa. M. S. “Indian Miniature Paintings”, Roli Books International, New Delhi 1981
2. Shrikant. Usha , “Designs for Lifetime”, PriyavratGoenka, The South Pacific Clothes Export Company Pvt. Ltd. Andheri West , Mumbai 2002
3. Dayal. Bharti, “Indian Art Series: Madhubani Art”, Niyogi Books, Noyogi Offset Pvt Ltd. New Delhi 2015
4. Lehri. M. S., “Indi-Folk: Folk and Tribal Designs of India” , Super Book House, Mumbai India, 2009
5. Satyawadi. S. “Unique Art of WarliPaintings”,D.K.Printworld (P) Ltd. ,New Delhi 2010
6. Taubes. F. “A Guide to Traditional and Modern Painting”, Murray Printing Co., Great Britania 1963

7. Goswamy B.N. “Pahari Painting of the Nala - Damyanti”, National Museum, New Delhi 1975
8. Randhawa. M.S. “Kangra Valley Painitng”, Ministry of Information and Broadcasting, Govt. Of India, Times of India, Bombay
9. Ebeling. K. “RagmalaPainitng”, Ravi Kumar, Basel, Paris, New Delhi 1973
10. Gillow. J. & Barnard N. “Indian Textiles”, Om Books International, New Delhi 2008
11. Hataraba. K. “Textile Parts of India” Cronicle Books, Kyoto Shoin Co. Ltd., Japan 1996
12. Aryan. K.C. “Basis of Decorative Elements in Indian art”, Rekha Prakashan, New Delhi 1981
13. Chishti. R.K.&Jain.R,“Hand Crafted Indian Textiles” Roli Books Pvt. Ltd.,Lusture Press, New Delhi, India, 2000
14. Mode.H. & Chandra. S., “Indian Folk Art”, Alpine Fine Arts Collection Ltd., New York 1985
15. “An Introduction to Indian Art”, Textbook in Fine Arts for Class XI, NCERT, New Delhi 2012
16. “Living Traditional Tribal & Folk painting of India”, Centre for Cultural Resources & Training, Ministry of Culture, Govt. of India, New Delhi
17. “Living Craft Tradition of India”, NCERT, New Delhi

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Home Science

Semester-I

Introduction to Applied Sciences for Home Science- I (Multi-disciplinary Course)
BSC-HSC-MDC-1-101
Theory

Credit: 1 **Maximum Marks: 25**
Duration of Class: 1hr/week **Paper: 20**
Duration of Exam: 3hrs **Internal Assessment: 05**

Instructions to the Examiner

- Each theory paper will be of two hours duration.
- Question paper will have four sections/units. Paper setter will set a total of nine questions comprising of two questions from each unit and one compulsory question of short answer type covering the whole syllabus.
- Student will attempt a total of five questions in all, selecting one question from each section/units including the compulsory question.
- All questions may carry equal marks, unless specified.

Learning Outcomes:

- To provide knowledge regarding the applications of science in day to day life.

Unit-I

1. Heat & Temperature: Definition & units, Applications/Illustrations of conduction, convection, Radiation in everyday life.
2. Principle, construction, working and Applications of Solar Cooker and Solar Panels.
3. Basics of Microwave oven, Air Fryer, Oven Toasted Griller.

Unit-II

4. Detergents and surface tension- cleansing action of Soaps & Detergents.
5. Types of Hard water, Heavy water and its uses.
6. Concept of Rain Water Harvesting.

Unit-III

7. Study of Different types of soil- Clay, sand, loam, Gravel, Alluvial Peat.
8. Vegetative propagation by artificial methods-Cutting, layering, grafting, budding.
9. Compost- Types and uses

Unit-IV

10. Principle and planning of kitchen garden.
11. Principle and planning in laying out of a garden/ Landscaping.
12. Hydroponics, Vertical Gardening Techniques.

Semester-I

Introduction to Applied Sciences for Home Science- I (Multi-disciplinary Course)
BSC-HSC-MDC-1-101
Practical

Credit: 2 **Maximum Marks: 50**
Duration of Class: 4hrs/week **Paper: 40**
Duration of Exam: 3hrs **Internal Assessment: 10**

Instructions to the Examiner

- The external and internal examiners are requested to be present at the examination center before the commencement of the examination.
- Set the question papers in consultation with the internal conduct the practical examination as per the scheme.

Learning Outcomes:

- To provide knowledge regarding the applications of science in day to day life.

Content

1. To Prepare a Kitchen Garden.
2. Study and identification of various Medicinal and Ornamental Plants.
3. Study of Garden implements (Garden Tools & accessories).
4. To prepare a pot for sowing seeds and study different seed sowing methods.
5. Propagation of roses by cutting.
6. Demonstration of conduction, convection, radiation.
7. Preparation of liquid soap
8. Preparation of washing powder

Recommended Readings

1. B. Choudhary: Vegetables (National Book of India, New Delhi 1979)
2. Breikell C. 1993, Step by Step Gardening Technique (Royal Horticultural Society's Encyclopedia of Practical Gardening).
3. Gopalaswamianger K.S. 1991 Complete Gardening in India (Messers Nagaraj and Co., Madras),H.T. Harman and D Keter: Plant Propagation, Principles and Practices (Prentice Hall of India Pvt. Ltd. New Delhi 1979).
4. Sh. S.K. Jain: Medicinal Plants: 2008 (National Book Trust, India).
5. Applied Chemistry for Home science and Allied science by Thancamma Jacob
6. NCERT books of +1 and +2.
7. Dutta A.C. Botany for Degree Students (Oxford University Press, New Delhi 1970)
8. Sudhir Pardhan: Economic Botany (Kitab Mahal, Allahabad 1982).
9. Sh. V. Verma : A text Book of Economic Botany 1984. Emkay Publication.
10. Gomber & Gogia, Pardeep's Fundamental Physics (XI, XII)
11. Gupta S.K, Modren's ABC of Physics (XI, XII)

Semester-II

Introduction to Applied Sciences for Home Science-II (Multi-disciplinary Course)
BSC-HSC-MDC-2-201
Theory

Credit: 1 **Maximum Marks: 25**
Duration of Class: 1hr/week **Paper: 20**
Duration of Exam: 3hrs **Internal Assessment: 05**

Instructions to the Examiner

- Each theory paper will be of two hours duration.
- Question paper will have four sections/units. Paper setter will set a total of nine questions comprising of two questions from each unit and one compulsory question of short answer type covering the whole syllabus.
- Student will attempt a total of five questions in all, selecting one question from each section/units including the compulsory question.
- All questions may carry equal marks, unless specified.

Learning Outcomes

- To provide knowledge regarding the applications of science in day to day life.

Unit-I

1. Applications of LASERS- Medical & Scientific, Industry & Commercial.
2. Safety Features in household electric wring of Fuse, MCB, Earthing.
3. Short note on Solar panels.

Unit II

4. Introduction to acids, bases & salts, - PH value of some common substances.
5. Importance of pH in digestive system, Tooth decay, Plant Growth
6. Cosmetics- Brief study & elementary idea about ingredients- cold cream, vanishing cream, lipstick, mascara, Food Additives.

Unit-III

7. Vermicomposting- Technology, Importance and Relevance.
8. Biofertilizers, Inorganic Fertilizers- Nitrogen Potassium & Phosphorus
9. Organic manures- Farm Yard manure, compost, Leaf mould manure, Green manure, Poultry & Pigeon manure, soot.

Unit-IV

10. A brief account of Principle of Blood flow & circulatory system dynamics.
11. Concept of A, B, O and Rh blood grouping.
12. Grain Storage: Principles of toxicology with reference to home.

Semester-II

Introduction to Applied Sciences for Home Science-II (Multi-disciplinary Course)
BSC-HSC-MDC-2-201
Practical

Credit: 2 **Maximum Marks: 50**
Duration of Class: 4hrs/week **Paper: 40**
Duration of Exam: 3hrs **Internal Assessment: 10**

Instructions to the Examiner

- The external and internal examiners are requested to be present at the examination center before the commencement of the examination.
- Set the question papers in consultation with the internal conduct the practical examination as per the scheme.

Learning Outcomes

1.) To provide knowledge regarding the applications of science in day to day life.

Content

1. To prepare a seed bed for raising seedlings.
2. To prepare a bed for sowing and cultivation of Potato.
3. To demonstrate propagation of plants by layering method-Air Layering, Ground Layering
4. Maintenance of plants – hoeing, mulching, weeding, staking & tieing.
5. Preparation of rose water, cold cream and vanishing cream
6. Preparation of antiseptic ointment
7. Preparation of fly/insect repellent solution
8. To identify vitamin C content in different fruits and vegetables through discolouration rate
9. To extract natural dyes from plants
10. Demonstration of measurement of pulse and blood pressure
11. Effect of exercise on pulse rate and blood pressure
12. Working/non-working models depicting scientific phenomena

Recommended Readings

1) Applied Chemistry for Home science and Allied science by Thancamma Jacob
2) NCERT books of +1 and +2.
3) Dutta A.C. Botany for Degree Students (Oxford University Press, New Delhi 1970)
4) Sudhir Pardhan: Economic Botany (Kitab Mahal, Allahabad 1982).
5) Sh. V. Verma : A text Book of Economic Botany 1984. Emkay Publication.
6) B.D. Hames & N.M. Hooper :Instant notes on Biochemistry
7) Dhami, PS: A text Book of Zoology for 11th & 12th students(2000)
8) Vidyarthy,R.D.:A text book of Zoology for intermediate students (1998).
9) Venkita Raman: A text book of Economic Zoology(1990)
10) Lal S.S.: Practical Zoology
11) Gomber & Gogia, Pardeep's Fundamental Physics (XI, XII)
12) Gupta S.K, Modren's ABC of Physics (XI, XII)

Semester – III

Economics of Entrepreneurship & Innovation (Multi-Disciplinary Course)
BSC-HSC-MDC-3-301
Theory

Credit: 2 **Maximum Marks: 50**
Duration of Class: 2hrs/week **Paper: 40**
Duration of Exam: 3hrs

Internal Assessment: 10

Instructions to the Examiner

- Each theory paper will be of three hours duration.
- Question paper will have four sections/units. Paper setter will set a total of nine questions comprising of two questions from each section and one compulsory question of short answer type covering the whole syllabus.
- Student will attempt one question from each unit and the compulsory question (Total of five questions)
- All questions may carry equal marks, unless specified.

Learning Outcomes

After studying the course, the students will be able to

- To develop an entrepreneurial mind-set, critical thinking, and problem-solving.
- To acquire the skills of innovation, creativity and management of entrepreneurial pursuits

Unit I

1. Entrepreneurial Economics- Need and importance. Entrepreneur and enterprise
2. Role of Entrepreneurship in Economic Development, Entrepreneurship as a Process, Barriers to entrepreneurship

Unit II

3. Definition, types and characteristics of entrepreneur. Functions and role of entrepreneur
4. Entrepreneurial Motivation and types. Factors affecting Entrepreneurial Motivation. Entrepreneurial Competencies and Creativity. Factors fostering competencies.

Unit III

5. Innovations and Intrapreneurship, Essentials for Innovation. Factors promoting intrapreneurship
6. Intrapreneurship Skills. Similarity and difference between Entrepreneurship and Intrapreneurship

Unit IV

7. Role of Technology – E-commerce and Social Media in promoting innovation and entrepreneurship
8. Social Entrepreneurship – Concept and types. Benefits of building a social enterprise

Recommended Readings

1. Gupta C.B and Srinivasan N.P, "Entrepreneurial Development" Sultan Chand and Sons, New Delhi
2. Khanna S.S, "Entrepreneurship Development" Sultan Chand and Sons, New Delhi
3. Patel, V. G., "The Seven Business Crises and How to Beat Them" Tata-McGraw, New Delhi, 1995.
4. SIDBI Report on Small Scale Industries Sector (Latest Editions)
5. Taneja, Satish and Gupta, S.L, "Entrepreneurship Development-New Venture Creating" Galgotia Publishing House, New Delhi, Latest Edition
6. Arora Renu and Sood, "Fundamentals of entrepreneurship and Small business" Kalyani publishers

Semester – III

Economics of Entrepreneurship & Innovation
BSC-HSC-MDC-3-301
Practical

Credit: 1 **Maximum Marks: 25**
Duration of Class: 2hrs/week **Paper: 20**
Duration of Exam: 3hrs **Internal Assessment: 05**

Instructions to the Examiner

- The external and internal examiners are requested to be present at the examination centre before the commencement of the examination.
- Set the question papers in consultation with the internal examiner and conduct the practical examination as per the scheme.

Learning Outcomes

(i) To familiarize the students with the process and skills of creation and management of enterprise.
(ii) To plan, use, monitor and control resources optimally and economically.

Content

1. Case study of small enterprises/cottage industries/startups in the respective domains of Home Science. Conducting first hand survey of the various aspects of respective enterprise including the process of setting up of an enterprise, motivating factors responsible for undertaking entrepreneurial set up , capital invested, production techniques adopted, marketing practices adopted for sustainability and future prospects of the enterprise surveyed
2. Project report on various institutions facilitating Entrepreneurship Development in India.
3. SWOC analysis of successful entrepreneurs and enterprises through case profiling
4. Developing a Business Plan for Micro enterprises. Calculation of Breakeven point and conducting feasibility analysis.

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Mathematics

NOTE: The following MDCs would be floated in each semester. However, the students are not allowed to repeat any paper.

BA/BSc (ANY SEMESTER)
MAT-MDC-101: BASIC MATHEMATICS

Credits = 3
Teaching hours = 5 hrs / week including tutorials / practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 75 (60 External + 15 internal assessment)

Note: 1. The syllabus has been split into three Units: Unit – I, Unit – II and Unit - III. The examiner will set in total seven questions including a compulsory question. The compulsory question (Question No.1) will contain 6 parts of 2 marks each, covering the entire syllabus. Two questions, each of 16 marks, will be set from each of the 3 Units.
2. A student will be required to attempt four questions in all, selecting one question from each Unit and the compulsory question.
3. 15 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 3 marks,
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Objective: The objective of this course is to teach basics of various topics in Mathematics which will make a strong foundation of students for their further learning in mathematics, physics, statistics etc

Learning outcomes: The course will enable the students to learn the following:
1. Set theory, Relations and Functions and Solution of Linear Inequalities in one and two variables.
2. Basics of sequences and series, in particular A.P. and G.P.
3. Basic Mathematical Reasoning.
4. Measures of central tendency and dispersion.

Unit –I

Sets: Definition, Types of Sets, Subsets, Operations on Sets, Venn Diagrams, Applications of Set Theory.
Relations and Functions : Definitions, Domain and Range, Graphs of standard functions, One-one and Onto functions, Inverse of a function.

Unit –II

Sequences and Series: Definitions, *n*th terms of A.P., G.P. and H.P. Sum of *n* terms of A.P. and G.P.
Mathematical Reasoning: Statement, Negation of a statement, Compound Statements, Connecting Words, Quantifiers, Implications, Validating Statements.

Unit – III

Statistics: Introduction, Measures of Central Tendency (Mean, Median, Partition values and Mode), Measure of Dispersion (Range, Quartile Deviation, Mean Deviation, variance and Standard Deviation), Coefficient of variation.

Essential Textbooks

1. Mathematics; Text book for class XI, NCERT.
2. Mathematics; Text book for class XII, NCERT (Part I and Part II).

BA/BSc (ANY SEMESTER)
MAT-MDC-102: CALCULUS

Credits = 3
Teaching hours = 5 hrs / week including tutorials / practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 75 (60 External + 15 internal assessment)

Note: 1. The syllabus has been split into three Units: Unit – I, Unit – II and Unit - III. The examiner will set in total seven questions including a compulsory question. The compulsory question (Question No.1) will contain 6 parts of 2 marks each, covering the entire syllabus. Two questions, each of 16 marks, will be set from each of the 3 Units.
2. A student will be required to attempt four questions in all, selecting one question from each Unit and the compulsory question.
3. 15 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 3 marks,
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Imp. Note: The question paper will be set from NCERT books only

Objective: The objective of this course is introduce trigonometry and basics of calculus, which includes limit, continuity and differentiability of real valued functions of real variables.

Learning outcomes: The course will enable the students to learn the following:
1. Trigonometric ratios and identities.
2. Limit of functions
3. Continuity and differentiability of functions.

Unit –I
Trigonometry: Trigonometric ratios (functions), Trigonometric identities, Sign of trigonometric ratios, Graph of trigonometric functions, Trigonometric ratios of sum and difference of two angles, Transformation of product into sum or difference of trigonometric functions, Transformation of sum or difference into product of trigonometric functions (i.e. C – D formulae), Trigonometric ratios of multiple angles.

Unit –II
Limit: Limit of a Function at a Point, Concept of Infinity, Algebra of Limits, Evaluation of Limits of Algebraic Functions, Limit using standard results such as

$$\lim_{x\to 0}\frac{\sin x}{x}=1,\ \lim_{x\to 0}\frac{e^x-1}{x}=1,\ \lim_{x\to 0}\frac{\log(1+x)}{x}=1,\ \lim_{x\to 0}(1+x)^{\frac{1}{x}}=e.$$

Continuity: Continuity of a function at a point, Types of Discontinuities, Algebra of Continuous Functions, Examples of Continuous Functions.

Unit – III
Differentiation: Derivative of Some standard functions (excluding trigonometric functions), Product rule for differentiation, Quotient rule for differentiation, Chain Rule, Derivative of implicit functions, Increasing and decreasing functions.

Essential Textbooks
1. Mathematics; Text book for class XI, NCERT.
2. Mathematics; Text book for class XII, NCERT (Part I and Part II).

BA/BSc (ANY SEMESTER)
MAT-MDC-103 : ALGEBRA

Credits = 3
Teaching hours = 5 hrs / week including tutorials / practicals etc. in a semester of 15 weeks.
Exam Time: 3 hours
Max. Marks: 75 (60 External + 15 internal assessment)

Note: 1. The syllabus has been split into three Units: Unit – I, Unit – II and Unit - III. The examiner will set in total seven questions including a compulsory question. The compulsory question (Question No.1) will contain 6 parts of 2 marks each, covering the entire syllabus. Two questions, each of 16 marks, will be set from each of the 3 Units.
2. A student will be required to attempt four questions in all, selecting one question from each Unit and the compulsory question.
3. 15 marks of internal assessment has following division
Mid Semester Exam = 10 marks,
Assignment / Project work = 3 marks,
Attendance = 2 marks for more than 85% and 1 mark for more than 75% attendance.

Objective: The objective of this course is introduce Principal of Mathematical Induction, Complex Numbers, Polynomials and Matrices.

Learning outcomes: The course will enable the students to learn the following:
1. Principal of Mathematical Induction
2. Complex numbers
3. Polynomials and Matrices

Unit – I
Principle of Mathematical Induction, Fundamental Principle of counting.
Linear Inequalities: Algebraic and Graphical Solution of Linear Inequalities in one and two variables.

Unit –II
Complex Numbers: Definition of a complex number, addition, subtraction, multiplication and conjugate of complex numbers. Representation of a complex number in a plane (Cartesian as well as polar coordinate plane), Modulus and Argument of a complex number.
Polynomials: Definition, Addition of polynomials, Product of polynomials, Euclid's algorithm, Remainder Theorem, Factor Theorem, Horner's Method of Synthetic Division (all theorems Without Proof).

Unit –III
Matrices: Definition, Types of Matrices, Equality of Matrices, Addition and Subtraction of Matrices, Scalar Multiplication, Multiplication of Matrices, Transpose of a Matrix, Symmetric and Skew-Symmetric Matrix.
Determinants: Determinant of Square Matrices of Order 1, 2 and 3. Application of determinants like area of a triangle, condition for collinearity of three points etc.

Essential Textbooks
1. **Mathematics**; Text book for class XI, NCERT.
2. **Mathematics**; Text book for class XII, NCERT (Part I and Part II).
3. **Mathematics**; Text book for class X, NCERT.

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Microbial Food Technology

SEMESTER-I

MULTI-DISCIPLINARY COURE (BMF-MDC-1-101)
NUTRITION AND DIETARY MANAGEMENT

(CREDITS-3+0)
TOTAL HOURS-30

MAX. MARKS:75
THEORY:60
INTERNAL ASSESSMENT:15

INSTRUCTIONS FOR THE EXAMINER: The Question Paper will have four sections. Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus. Students will attempt one question from each unit and the compulsory question. All questions may carry equal marks, unless specified.

COURSE OBJECTIVES: To understand the scope of specialty foods. To classify specialty foods on basis of sources. To familiarize with the concept of therapeutic foods. To gain insights into consumer requirements in area of specialty foods.

LEARNING OUTCOMES: The students will understand the scope and significance of specialty foods in the food industry.They will be able to learn about specialty foods based on their sources and unique characteristics, concept and applications of therapeutic foods and the trends related to specialty and functional foods.

UNIT I **No. of Hours: 07**

Need and Scope of Specialty Foods: Need and scope of specialty foods, Functional foods, Convenience food, Health care and medical benefits, Nutritional status, Low-cost foods. Nutritional labelling.

Specialty Foods Based on Sources: Cereals and millets, Legumes and pulses, Fruits and vegetables, Animal food sources, By-product based, Bioactive components

UNIT II **No. of Hours: 08**

Introduction to Human Nutrition: Introduction to human nutrition, energy value of foods and from proximate principles, daily calorie needs for basal metabolism.

Dietary Requirements of Nutrients: Requirements and role of carbohydrates, lipids, water, proteins, vitamins, and minerals in human health, recommended dietary allowance (RDA), dietary sources. anti-nutritional factors in plant foods.

UNIT III **No. of Hours: 08**

Therapeutic Foods: Modification of diets in disorders of digestive tract, liver, cardiovascular system, kidney, metabolic disorders, allergy, endocrine disorders.

Specific Consumer Oriented Foods: Defence persons, Space / astronaut, High altitude mountain climbers, Disaster situation – crises, care, and maintenance

UNIT IV **No. of Hours: 07**

Nutrition Management Techniques: Infant nutrition and infant foods, geriatric nutrition, maternal nutrition, Sports Nutrition.

Advances in Nutrition: Effect of processing, preservation, and storage on nutritional quality of foods, food fortification, food enrichment, complementation and supplementation diets.

SUGGESTED READINGS (latest editions)

1. Gibson GR & William CM. 2000. Functional Foods - Concept to Product.
2. Robert EC. 2006. Handbook of Nutraceuticals and Functional Foods. 2nd Ed. Wildman.
3. Manson P.2001. Dietary Supplements. 2nd Ed. Pharmaceutical Press.
4. Bamji MS, Rao NP & Reddy V. 2003. Textbook of Human Nutrition. Oxford & IBH.
5. B Srilakshmi. 2014. Dietetics 8thed. New Age Publication

SEMESTER-II

MULTI-DISCIPLINARY COURE (BMF-MDC-2-201)

DESIGNER FOODS AND HEALTH

(CREDITS-3+0)
TOTAL HOURS-30

MAX. MARKS: 75
THEORY: 60
INTERNAL ASSESSMENT: 15

INSTRUCTIONS FOR THE EXAMINER: The Question Paper will have four sections. Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus. Students will attempt one question from each unit and the compulsory question. All questions may carry equal marks, unless specified.

COURSE OBJECTIVES: The course is designed to inculcate the knowledge about the functional foods, their importance and health benefits. Further, the students will acquire knowledge on the health promoting nutritional factors and bioactive constituents, their potential health implications and mechanisms of action.

LEARNING OUTCOMES:Students will understand the concept of functional foods and their importance in promoting health.They will learn about the health benefits of bioactive constituents and their mechanisms of action.Students will gain knowledge on the nutritional factors that contribute to health promotion.

UNIT I **No. of Hours: 08**

Functional Foods and Nutraceuticals and Role in Health Management: Introduction to Functional Foods: Definition, History and Classification; Introduction to Nutraceuticals: Definition, Health benefits; Applications of Nutraceuticals in health and wellness; Introduction to Probiotics, Prebiotics and Synbiotics.

UNIT II **No. of Hours: 07**

Probiotics, Prebiotics and Synbiotics: Probiotics-Taxonomy and Important Features of Probiotic Microorganisms; Quality Assurance of Probiotics and Safety; Probiotics in various foods and health benefits; Types of prebiotics and their health benefits

UNIT III **No. of Hours: 07**

Functional foods and their types: Conventional functional foods (e.g Millets); Modified functional foods (Fortification, Enrichment and supplementation); Foods for special dietary use (Weight loss foods, Gluten free foods, Lactose free foods)

UNIT IV **No. of Hours: 08**

Bioactive Components in Foods and Health effects: Introduction to bioactive compounds: description, health effects, and food examples; Polyphenols, flavonoids, carotenoids; Anti-nutritional Factors, Phytates; Vitamins, Minerals; Special reference to Resveratrol, Quercetin, Cinnamaldehyde, Capsaicin.

SUGGESTED READINGS (latest editions):

1. Wildman, R. E. (2016). Handbook of Nutraceuticals and Functional Foods. CRC Press
2. Gibson, G. R. and Williams, M. C. (2001). Functional Foods Concept to Product. CRC Press.
3. Vattem, D.A. and Maitin V.(2016). Functional Foods, Nutraceuticals and Natural Products, Concepts and Applications. DEStech Publications, Inc
4. Gupta, R. C. (2016). Nutraceuticals: Efficacy, Safety and Toxicity. Academic Press.
5. Robert E.C. Wildman; Handbook of Nutraceuticals and Functional Foods, Second Edition; CRC Press
6. Giuseppe Mazza; Functional Foods: Biochemical and Processing Aspects, Volume 1; CRC Press

SEMESTER-III

MULTI-DISCIPLINARY COURE (BMF-MDC-3-301)
COMMUNITY NUTRITION

(CREDITS-3+0)
TOTAL HOURS-30

MAX. MARKS:75
THEORY:60
INTERNAL ASSESSMENT:15

INSTRUCTIONS FOR THE EXAMINER: The Question Paper will have four sections. Examiner will set a total of nine questions comprising two questions from each unit, and one compulsory question of short answer type covering the whole syllabus. Students will attempt one question from each unit and the compulsory question. All questions may carry equal marks, unless specified.

COURSE OBJECTIVES: To provide information regarding nutritional assessment.
- To enable students to impart nutrition education among rural and needy people.
- To acquaint them knowledge regarding food security and government and international program running in the field of community nutrition.

LEARNING OUTCOMES: Students will learn methods of nutritional assessment and their application in community settings.They will develop the ability to plan and deliver effective nutrition education to rural and underserved populations.Students will gain knowledge of food security concepts and challenges.They will understand the role of government and international programs in promoting community nutrition.

UNIT I — No. of Hours: 07

Introduction: Nutrition and health in National development, food consumption patterns and choices, food security and poverty.
Malnutrition: meaning. factors contributing to malnutrition, over nutrition.

UNIT II — No. of Hours: 08

Nutritional disorders: prevention and dietary treatment for Protein Energy malnutrition, nutritional anemias& vitamin deficiency disorders.
Methods of assessing nutritional status: a) Sampling techniques, Identifications of risk groups, Diet surveys, Food balance sheet and ecological parameters

UNIT III — No. of Hours: 08

Improvement of nutrition of a community: Modern methods of improvement or nutritional quality of food, food fortification, enrichment, and nutrient supplementations.
Nutritional and infection relationship: Immunization and its importance, Overview of Food borne diseases and control measures.

UNIT IV **No. of Hours: 07**

National and International agencies in uplifting the nutritional status: WHO, UNICEF, CARE, ICMR, ICAR, CSIR, CFTRI.

Community nutrition program planning : Identification of problem, analysis of causes, setting a strategy, implementations, and evaluation of the program.

SUGGESTED READINGS (latest editions)

1. Bamji,M.S Rao, NP and ReddyV. 1996.Text book of Human Nutrition
2. Bagchi, K. 1990. Guidelines for the management of nutrition programmes- a manual for nutrition officers. WHO EMRO Technical Publication no. 15, WHO,Geneva
3.Bendich, A and Deckelbaum,RJ, 1997. Preventive Nutrition. The Comprehensive guide for health professional.

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Microbiology

MIC-MDC-1-101: INTRODUCTION TO MICROBIOLOGY

Total marks: 50
Theory: 45
Internal assessment: 05
Time: 3 hours
Credits: 02

Note: The question paper will consist of Two Units. There will be nine questions, out of which five questions have to be attempted. Question I will span the complete syllabus and will be compulsory. Rest of the eight questions will be from different Units of the syllabus. There will be four questions from each Unit and two is to be attempted from each unit

Objective And Learning outcome: Aim of the course is to introduce the students with the microbes .Microbes play a significant role in medical science and industries .To study the microbes from basic, with an understanding of cell structure, and application of microbes.

UNIT–I

HISTORY AND SCOPE OF MICROBIOLOGY
General introduction and overview of microbial world. Major division of life. Members of microbial world and their characteristics. The Morphology of Bacteria, viruses, viroids and Prions, fungi, Algae, protozoa. The development of techniques for studying microbial pathogens. The scope and relevance of microbiology.

MICROBIOTA AND HOST RELATIONSHIP
The distribution and role of Normal microflora of human Body. The role of microorganisms in diseases. Microbial interactions like symbiosis and antibiosis.

UNIT-II

MICROSCOPY
Principles of microscopy; resolution, magnification; compound microscope**,** Bright field microscopy, Dark field microscopy. The influence of environmental factors on growth of microorganisms.

MEDIA PREPARATION
Culture media: Different types of media preparation methods. Sterilization and pasteurization; The Physical and chemical agents to kill microbes.

SUGGESTED READING

1. Willey, J. M., Sherwood, L. M., & Woolverton, C. J. (2014). *Prescott's microbiology*. McGraw-Hill.
2. Black JG. (2008). Microbiology: Principles and Explorations. 7th edition. Prentice Hall
3. Pelczar Jr MJ, Chan ECS, and Krieg NR. (2004). Microbiology. 5th edition Tata McGraw Hill.
4. Srivastava S and Srivastava PS. (2003). Understanding Bacteria. Kluwer Academic Publishers, Dordrecht.
5. Stanier RY,Ingraham JL, Wheelis ML and Painter PR. (2005). General Microbiology. 5th edition McMillan.
6. Tortora, G.J., Funke, B.R. and Case, C.L. (2009) Microbiology: An introduction (Benjamin/ Cummings publishing company, Inc).

7. Stanier R Y, M. Doudoroff, E. A. Adelberg (1999). General microbiology (MacMillian press London).
8. Willey J M, Sherwood LM, and Woolverton CJ. (2013). Prescott's Microbiology. 9th edition. McGraw Hill Higher Education.
9. Postgate, J.R. (2000). Microbes and man (Cambridge University Press).
10. Alcamo, I. E. (2003). *Microbes and society: an introduction to microbiology*. Jones & Bartlett Learning.
11. Cappucino J and Sherman N.(2010).Microbiology: A Laboratory Manual9th edition. Pearson Publication

MIC-MDC-1-101: INTRODUCTION TO MICROBIOLOGY (PRACTICAL)

Total: 25
Practical: 22
INTERNAL ASSESSMENT:03
TIME: 3 hours
CREDITS:1

List of the Practical:

1. Basic Laboratory Rules and Instruments used in microbiology lab.
2. The components, use, care and measurements of microorganisms of the compound Microscope.
3. Sterilization of items and Preparation of culture media.
4. Microscopic examination of microorganism by Temporary Wet Mount.
5. Demonstration of Motility by hanging drop method.
6. Demonstration of Normal Microbial flora of the skin.
7. Dilution and pour plating techniques.

Semester-II

MIC-MDC-2-201: TOOLS AND TECHNIQUES IN MICROBIOLOGY (THEORY)

Total marks: 50
Theory: 45
Internal assessment: 05
Time: 3 hours
Credits: 02

Objective: To enable the students learn important tools and techniques in microbiology so that they can understand application of these techniques.

Unit I

Principle, working and applications of Spectrophotometry (UV & visible)
Electrophoresis: Principle, types & applications.

Unit II

Microscopy: Properties; Light and Bright field Microscopy, Dark field Microscopy; Phase contrast, Fluorescent microscopy, Electron microscopy: Scanning and Transmission Electron Microscopy.

Unit III

Overview of chromatography; ion exchange chromatography, gel filtration chromatography

Unit IV

Radioisotope techniques: Radiotracers; GM counter, proportional and scintillation counters, autoradiography

SUGGESTED READINGS

1. Willey, J. M., Sherwood, L. M., & Woolverton, C. J.(2014). Prescott's microbiology. McGraw-Hill.
2. Wilson, K., Hofmann, A., Walker, J. M., & Clokie, S. (Eds.). (2018). Wilson and Walker's principles and techniques of biochemistry and molecular biology. Cambridge university press.
3. Pommerville J.C. (2011). Alcamo's Laboratory Fundamentals of Microbiology. (Jones & Bartlett Publishers, London).
4. Singh B.D. (2008). Biotechnology: Expanding Horizons. (Kalyani Publishers, India).
5. Patel A.H. (2007). Industrial microbiology. (New Age International Publishers).
6. *Aneja, K. R. (2007). Experiments in microbiology, plant pathology and biotechnology. New Age International.*
7. Alcamo, I. E. (2003). *Microbes and society: an introduction to microbiology*. Jones & Bartlett Learning.
8. Cappucino J and Sherman N.(2010).Microbiology: A Laboratory Manual 9th edition. Pearson Publication

MIC-MDC-2-201: TOOLS AND TECHNIQUES IN MICROBIOLOGY (PRACTICAL)

Total marks:25
Practical:22
Internal assessment:03
Time:3 hours
Credits: 01

List of Experiments:

1. Quantitative analysis by UV/ Visible spectrophotometry.
2. Agarose gel electrophoresis
3. Paper chromatography
4. Gel filtration chromatography
5. Parts and working of microscope

MIC-MDC-3-301: ENTREPRENEURSHIP IN MICROBIAL BIOTECHNOLOGY (THEORY)

Total Credit 2
Total marks 50
Internal assessment 5
Theory exam: 45

Course Objective:

To impart knowledge on microbial biotechnology with a focus on entrepreneurial opportunities, innovation, and the development of viable business ventures using microbial products and processes.

Instructions for examiners

Set nine questions in total. Question 1 is compulsory and cover whole syllabus and should be very short answer type. Set two questions from each unit. One question to be attempted in each unit. All questions carry equal marks.

Unit -I

Introduction to microbial biotechnology: Scope and significance, Overview of microbes in industrial and environmental processes, Concepts and characteristics of entrepreneurship, Entrepreneurial motivation and competencies, Concept of bio-entrepreneurship, significance of bio-entrepreneurship & their advantages and disadvantages, essential bio-entrepreneur characteristics, bio-entrepreneurship versus general entrepreneurship

Unit -II

Innovation as a strategy in bio-entrepreneurship, Assessment of innovative ideas, acquisition of novel ideas, product assessment for entrepreneurship Government initiatives for startups: Biotech Parks, technology business incubators, special economic zones.

Microbiology-based entrepreneurship: Recent developments in commercializing important microbial products such as: enzymes; novel biomolecules of therapeutic use; primary metabolites (ethanol, citric acid); secondary metabolites (beta exotoxin); single cell proteins (spirullina, chlorella), yeast (candida); mushroom, probiotics.

Unit –III

Steps for initiating a start-up: Ideation to execution, Infrastructure and regulatory requirements (FSSAI, DBT, CPCB, MSME), Technology transfer and Intellectual Property Rights (IPR) in biotech, Quality control and biosafety norms, Business model canvas for microbial enterprises.

Unit -IV

Sources of funding: Government schemes (BIRAC, DBT-BioNEST, Startup India), angel investors, Writing a biotech business plan and pitching ideas, Branding, product positioning, and digital marketing, Challenges in scaling microbial processes, Sustainable practices and future prospects.

SUGGESTED READINGS

1. T. Satyanarayana et al., Microbial Biotechnology: Fundamentals of Applied Microbiology, Springer, 2014.
2. R. C. Dubey, A Textbook of Biotechnology, S. Chand Publishing.
3. S.S. Khandare & P. Khandare, Entrepreneurship Development and Communication Skills, S. Chand.
4. U. Satyanarayana, Biotechnology, Elsevier India.

MIC-MDC-3-301: ENTREPRENEURSHIP IN MICROBIAL BIOTECHNOLOGY (PRACTICAL)

Total Credit: 1
Total marks: 25
Internal assessment: 3
Practical Exam : 22

1. Cultivation and quantification of microbial biomass for product development.
2. Screening of microbial isolates for enzyme (amylase, protease) production.
3. Preparation of prototype: Microbial bio-fertilizer or bio- pesticide formulation.
4. Designing a Business Model Canvas for a microbial product.
5. Visit to biotech incubator or interaction with a startup founder.

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Physics

B.Sc. Semester 1
Multidisciplinary Course (PHY-MDC-1-101)
(Green Energy)

Maximum marks: 75
(Theory 60 marks + Internal Assessment 15 marks)
Teaching hours: 3Hrs per week
Total Credits: 3

Instructions:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.

Learning Outcomes: The course provides a comprehensive understanding of work, energy, and power fundamentals. It includes various forms of energy such as renewable and conventional systems like coal, oil and natural gas. It explores the impact of non-conventional energy sources on global warming and examines approaches to energy conservation and governmental policies. Moreover, this course also covers specific renewable energy sources like solar, wind, hydro, tidal, and wave energy.

UNIT – I

INTRODUCTION: Fundamentals of Work, Energy and Power, Various Forms of Energy - Renewable and Conventional energy Systems, Comparison: Coal, Oil and Natural Gas– Availability, Applications, Merits and demerits, Impact Due to Non-Conventional Energy Sources – Global Warming.

SOLAR ENERGY: Solar Radiation Measurements (qualitative only), Solar Energy Collector, Principles of the Conversion of Solar Radiation into Heat, Classification of Different Types of Solar Energy Collectors (qualitative ideas only)- Merits and Demerits, Solar Energy Storage, Solar Heaters, Solar Cookers, Solar Green Houses, Merits and Demerits of Solar Energy.

UNIT – II

WIND ENERGY: Basic Principles of Wind Energy Conversion, Merits and Demerits, Basic Components of Wind Energy Conversion System, Policies Related to Wind Energy in India, Applications of Wind Energy.

HYDRO, TIDAL AND WAVE ENERGY: Hydro-Resources, Hydro-Project- Types and Hydro Conversion Technologies, Tidal Resource, Tidal Power Conversion, Wave Resource, Wave Energy Conversion, Challenges to Sustainability.

BOOKS:

1. Non-Conventional Energy Resources: G. D. Rai, Khanna Publishers, 2008.
2. Energy Technology: S. Rao and Dr. B.B. Parulekar, Third edition, 2009.
3. Alternative Energy Resources, Green Energy and Technology, Efstathios E. (Stathis) Michaelides, Springer, 2012, DOI 10.1007/978-3-642-20951-2.
4. Solar Energy Fundamentals and application: H.P. Garg and J. Prakash, Tata McGraw - Hill Publishing company Ltd., 1997.

B.Sc. Semester-II

Multidisciplinary Course (PHY-MDC-2-201)
Dynamics of Motion and Human Health

Maximum marks: 75
(Theory 60 marks+ Internal Assessment 15 marks)
Teaching hours: 3Hrs per week
Total Credits: 3

Instructions:
1. The theory paper will be of 75 marks including 15 marks for internal assessment and 3-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions from Unit I and II and the compulsory question from Unit III.
3. Use of non-programmable calculators will be allowed in the examination centre.

Learning Outcomes: Students will understand fundamental concepts in measurement, motion, forces, and energy, and apply them to real-life situations like sports, health, and nutrition. They will analyze the laws of physics in everyday contexts such as cycling, swimming, and diet, and appreciate the role of physical activity and balanced nutrition in maintaining overall well-being.

UNIT-I

Measurement: Physical quantities. Standards and Units. International System of Units. Standards of time, length and mass. Precision and significant figures.
Newton's laws of motion: Newton's firstlaw. Force, mass. Newton's second law. Newton's third law. Mass and weight. Applications of Newton's laws.
Projectile motion: Shooting a falling target. Physics behind Shooting, Javelin throw and Discus throw.
Conservation laws: Conservation of linear momentum, collisions- elastic and inelastic Angular momentum. (Physics behind Carom, Billiards, Racing)

UNIT-II

Centre of mass: Physics behind Cycling, rock climbing, Skating, Gravitation: Origin, Newton's law of gravitation. Archimedes' principle, Buoyancy (Physics behind swimming)
Food andNutrition: Proteins, Vitamins, Fat, Blood Pressure. Problems due to the deficiency of vitamins. Energy: Different forms of Energy, Conservation of mass-energy.
Physical exercises: Walking, Jogging and Running, Weight management.

Suggested Books
1. Physics for Entertainment-Yakov Perelman, Createspace Independent Pub
2. Physics Everywhere, Yakov Perelman - Prodinnova
3. Mechanics for Entertainment- Yakov Perelman-Prodinnova
4. Food Science- Sri Lakshami, New Age Publications
5. Physics, Resnick, Holiday and Krane, Wiley Student Edition
6. An introduction to the Physics of Sports-Vassilios McInnes Spathopoulos, Createspace Independent publishing Platform

B.Sc. Semester-III

Multidisciplinary Course (PHY-MDC-3-301)
(Basic Computational Tools)

Maximum marks: 75
(Theory 60 marks+ Internal Assessment 15 marks)
Teaching hours: 3Hrs per week
Total Credits: 3

Instructions:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and 3-Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions from Unit I and II and the compulsory question from Unit III.
3. Use of non-programmable calculators will be allowed in the examination centre.

Learning Outcomes: The objective of this course is to understand scientific computing fundamentals, including numerical methods, floating-point arithmetic, precision, and iterative techniques. To master C/C++ programming basics: variables, data types, control structures, functions, and object-oriented concepts. It enables us to solve practical problems like sorting, searching, random number generation, mathematical equations and apply programming to real-world scenarios, such as radioactive decay and classical motion equations.

Unit I

Basics of scientific computing: Binary and decimal arithmetic, Floating point numbers, algorithms, Sequence, Selection and Repetition, single and double precision arithmetic, underflow & overflow-emphasize the importance of making equations in terms of dimensionless variables, Iterative methods. Review of C & C++ Programming: Introduction to Programming, constants, variables fundamentals and data types, operators and Expressions, I/O statements, scanf and printf, c in and c out, Manipulators for data formatting, Control statements (decision making and looping statements) (If statement. If else Statement,Nested if Structure, Else-if Statement, Ternary Operator, Go to Statement, Switch Statement, Unconditional and Conditional Looping, While Loop, Do-While Loop, FOR Loop. Break and Continue Statements. Nested Loops), Arrays and strings, user defined functions, Structures and Unions, Idea of classes and objects.

Unit II

Programs: Sum & average of a list of numbers, largest of a given list of numbers and its location in the list, sorting of numbers in ascending descending order, Binary search Random number generation, Area of circle, area of square, volume of sphere, value of pi (π), Solution of linear and quadratic equation. Solution of first order differential equation, Radioactive decay, Classical equations of motion.

Books:

1. Schaum's Outline of Programming with C++. J. Hubbard, 2000, McGraw□Hill Pub.
2. Numerical Recipes in C: The Art of Scientific Computing, W.H. Pressetal, 3rdEdn. , 2007, Cambridge University Press.
3. An Introduction to Computational Physics, T. Pang, 2ndEdn., 2006,Cambridge Univ. Press.
4. Computational Physics, Darren Walker, 1stEdn., 2015, Scientific International Pvt. Ltd.

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STATISTICS

SEMESTER- I

Course Code: (STAT-MDC-1-101)
Course Name: Statistics-I

Credit: 3
L-3

Notes:

1. This course will be offered by the Department of Statistics to the students of other Departments (other than the Department of Statistics)
2. Students are to be engaged to 3 hours of teaching per week.

Paper Title with code	Lecture(L)/ Credit	Marks Distribution
Statistics-I (STAT-MDC-1-101)	Lecture : 3 Credit : 3	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours

STATISTICS-I

Objectives

This course is designed to shape the attitudes of learners regarding the field of statistics**.** This course is aimed at students who need a basic background in statistics and its application. It is to provide an understanding for the student on statistical concepts to include measurements of location and dispersion, probability, probability distributions, regression, and correlation analysis. It provides knowledge about basic probability theory, distributions and its significance in the real world. To analyze univariate and bivariate data, discrete and continuous random variables.

Learning Outcomes: On completion of this course students will be able to:

1. Understand the importance and scope of Statistics.
2. Get familiar with various techniques used in summarization and analysis of data.
3. Understand the nature of statistical data.
4. Understand about the types of correlations for bivariate and multivariate data and their uses in real life problems.
5. Highlight the concept of linear regression and its applications in real life
6. Understand the significance of statistics and probability in the real world

Note: There will be in all nine (9) questions, all of equal marks. The first question is compulsory and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each Unit. The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each Unit.

Unit -I

Introduction: Definition, importance and scope of statistics, Types of statistical data: primary and secondary data, qualitative and quantitative data, time series data, discrete and continuous data, Scales of measurement - nominal, ordinal, interval and ratio, Frequency distributions, cumulative frequency distributions, Diagrammatic representation of data: Bar diagrams, histogram, pie chart, measures of central tendency, Measures of dispersion, moments, skewness, kurtosis, Bivariate data: Definition, scatter diagram, Simple Correlation coefficient and its properties, rank correlation, Simple linear regression, properties, Fitting of linear regression equations using principle of least squares, partial and multiple correlation coefficient in three variables (only results no derivation).

Unit-II

Basic concepts of probability: Random experiment, sample space, events, different definitions of probability, Additive law of probability, multiplication probability, conditional probability, Random variables(for univariate case only): discrete and continuous random variables, Probability density function, distribution functions, mathematical expectation, moment generating function,

SUGGESTED READINGS:

1. Goon, A.M., Gupta, M.K. and Dasgupta, B. (2013): Fundamental of Statistics, Vol I , World Press, Kolkata
2. Miller, Irwin and Miller, Marylees (2006): John E. Freund's Mathematical Statistics with Applications, (7th Edn.), Pearson Education, Asia.
3. Gupta, S.C. &Kapoor, V.K. : Fundamentals of Mathematical Statistics, Sultan Chand and Sons.
4. Mood, A.M. Graybill, F.A. and Boes, D.C. (2011): Introduction to the Theory of Statistics, 3rd Edn., (Indian Edition), Tata McGraw-Hill Pub. Co. Ltd.

SEMESTER-II

Course Code: (STAT-MDC-2-201)
Course Name: Statistics-II

Credit: 3
L-3

Notes:

1. This course will be offered by the Department of Statistics to the students of other Departments (other than the Department of Statistics)
2. Students are to be engaged to 3 hours of teaching per week.

Paper Title with code	Lecture(L)/Credit	Marks Distribution
STATISTICS-II (STAT-MDC-2-201)	Lecture : 3 Credit : 3	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours

STATISTICS-II

Objectives:

The purpose of this course is to introduce students to the application of statistical methods and to make inferences about populations. Among the statistical methods that students will be exposed to are the following: t-tests, F- distribution, Chi-square distribution and one-way and two-way ANOVA. It provides knowledge to perform hypothesis tests on means and proportions for one or two populations for small as well as large sample.

Learning Outcomes: On successful completion of the course, Students will be able to:

- Understand concepts of sample, population, statistical hypotheses and their significance in real world.
- Analysis and also perform tests based on t, F, chi square and normal variate z for one and two samples.
- Perform one way and two way analysis of variance.
- Form test of hypotheses as well as calculate confidence interval for a population parameter for single sample and two sample cases.

Note: There will be in all nine (9) questions, all of equal marks. The first question is compulsory and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each Unit. The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each Unit.

Unit-I

Concept of population and sample, parameter and statistic, need of sampling, types of sampling (definition only), sampling distribution of a statistic, standard error, Statistical hypothesis, null and alternative hypotheses, simple and composite hypotheses, procedure in hypothesis testing, types of errors, power of test and critical region, levels of significance, one and two tailed test, degrees of freedom, Tests of hypothesis-large sample tests: test of hypothesis for single mean, difference of means, proportions, difference of proportions.

Unit-II

Sampling distributions and their applications: Student - t distributions, F- distribution and Chi-square distribution. Simple tests based on t, F, chi square and normal variate z. Analysis of variance: for one-way classification, two-way classification (for fixed effect model only).

SUGGESTED READINGS:

1. Hogg, R.V., Tanis, E.A. and Rao J.M. (2009): Probability and Statistical Inference, Seventh Ed, Pearson Education, New Delhi.
2. Gupta, S.C. &Kapoor, V.K. : Fundamentals of Applied Statistics, Sultan Chand and Sons.
3. Goon, A.M., Gupta M.K. &Dasgupta, B. (2005) : Fundamentals of Statistics, Vol. II, Word Press Calcutta.
4. Meyer, P.L. (1970): Introductory Probability and Statistical Applications, Oxford & IBH Publishing, New Delhi .

SEMESTER- III

Course Code: (STAT-MDC-3-301)
Course Name: Applied Statistics

Credit: 3
L-3

Notes:

1. This course will be offered by the Department of Statistics to the students of other Departments (other than the Department of Statistics)
2. Students are to be engaged to 3 hours of teaching per week.

Paper Title with code	Lecture(L)/Credit	Marks Distribution
Applied Statistics (STAT-MDC-3-301)	Lecture : 3 Credit : 3	Theory Max. Marks: 75 Theory: 60 Internal Assessment : 15 Time: 3 Hours

APPLIED STATISTICS

Objectives: The Learning Objectives of this course are as follows:

1. Introduce the concept of time series, its components, and their estimation.
2. Introduce the application of time series.
3. Introduce the concept, formulation, and application of index numbers.
4. Provide knowledge about different measures of mortality, fertility and population growth.

Learning outcomes: After completion of this course, the students will be able to:

1. Understand the concepts of time series and index numbers.
2. Formulate, solve, and analyse the use of time series and index numbers for real-world problems.
3. Understand and apply different measures of fertility, mortality and population growth to analyse the demographic data.
4. Apply index number and time series concepts for making future planning and policy related to business.

Notes: There will be in all nine (9) questions, all of equal marks. The first question is compulsory and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each Unit. The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each Unit.

UNIT-I

Index Number: Definition, problems involved in the construction of index numbers, calculation of index numbers-simple aggregate method, weighted aggregative methods, Laspeyre's, Paasche's, Marshall-Edgeworth and Fisher's index numbers, time and factor reversal tests of index numbers, simple average of price relatives, weighted average of price relatives, Chain base index numbers, value index numbers, price and quantity index numbers, consumer price index number, uses and limitations of index numbers.

UNIT-II

Demographic methods: Sources of demographic data-census, register, adhoc survey, hospital records, measurement of mortality, crude death rate, specific death rate, standardized death rates, measurement of fertility-crude birth rate, general fertility rate, specific fertility rate, total fertility rate, measurement of population growth, gross reproduction rate, net reproduction rate

Time Series Analysis: Definition, components of time series-trend, seasonal variations, cyclic variations, irregular component, illustrations, additive and multiplicative models, determination of trend: graphic method, semi-averages method, method of curve fitting by principle of least squares,moving average method.

Books recommended

S. No.	Title of Book	Name of author	Publisher
1.	Applied General Statistics	Croxton F.E., Cowden D.J. &Kelin S.	Prentice Hall
2.	Demography	Cox P.R.	Cambridge Uni. Press
3.	Technical Demography	Ramakumar R.	New Age International
4.	Fundamentals of Applied Statistics	Gupta S.C. & Kapoor V.K.	Sultan Chand & Sons

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Zoology

Course Code: ZOO-MDC-1-101
Course Title: Health and Hygiene

Total Hours= 45 (Theory=45) **Credits= 3 (L=3 + P=0)**
Total Marks= 75 (Theory=75)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Duration for final examination: 3 hours

Note: - This is a three credit introductory course. It is not the same as the courses studied by the students at the 12th level. It does not belong to the core Zoology. It is an interdisciplinary course and can be useful for all the disciplines of Science, Humanities, Commerce, etc. The syllabus can be taught by teachers from any discipline who knows the subject. Therefore, the course may be offered to students of all disciplines irrespective of the subjects studied at the 12th level.

Theory

Learning Outcomes:

- *The students will be able to develop clarity on key concepts related to health and hygiene.*
- *The students will able to develop strategies for the promotion of health and hygiene.*
- *The students will be able to appreciate different dimensions of health and role of proper nutrition.*
- *The students will be able to understand the importance of hygiene, personal and community health.*

UNIT 1

Nutrition

- Classification of foods -Carbohydrates, proteins, lipids, vitamins and minerals.
- Balanced diet and Malnutrition.
- Nutritional deficiencies and disorders.

UNIT 2

Health

- Definition of Health.
- Concept of Health & Diseases.
- Factors influencing Health.
- Immunity: Innate and Acquired Immunity.
- Basic concept of Vaccines and Allergy.

UNIT 3

Diseases

- Introduction to diseases, types of diseases: Communicable and Non-Communicable diseases.
- Communicable Diseases: Causes, Symptoms and Prevention - Measles, polio, Influenza, Chikungunya, Rabies and AIDS.
- Non-Communicable diseases and their preventive measures – Hypertension, Coronary Heart diseases, Stroke and Diabetes.

UNIT 4

Hygiene

- Hygiene: Concept, Meaning, Principles, and Importance.
- Standard Hygiene Practices.
- Types of Hygiene: Personal, Food, and Community.
- Health awareness, Food adulteration and inspections.

SUGGESTED READINGS

1. All India Institute of Hygiene and Public Health: http://aiihph.gov.in/healthin-india-2016/Page 5 of 10
2. Dunn, C. L. &Pandaya, D. D. (1925). Indian Hygiene and Public Health. Calcutta: Butter Worth Co. (India) Ltd.
3. Ministry of Ayush: http://ayush.gov.in/ Ministry of Health and Family Welfare: https://mohfw.gov.in/
4. Park, K.(2009).Preventive and Social Medicine. Jabalpur: M/s Banarsi das Bhanot.
5. Park, W. H. (2019). Public Health and Hygiene: In Contributions by Eminent Authorities Germany: Forgotten Books.
6. Sharma, M. Branscum, P. &Atri, A. (2014). Introduction to Community and Public Health. USA: John Wiley & Sons.
7. Swachh Bharat Mission: http://swachhbharatmission.gov.in/sbmcms/index.htm
8. KeshavSwarnkar, Community health –NR Brothers Publishers Indore
9. JatinV.Modi and Renjith S. Chawan. Essentials of Public Health and Sanitation–Part I-IV
10. Murray, C. J. L. and A.D. Lopez. (1996). The Global Burden of Disease. World Health Organization. Park, J.E. and Park, K. Textbook of Community Health for Nurses.
11. Swaminathan S. Principles of Nutrition and Dietetics.
12. Personal Hygiene- objectives and advantages – Health and optimum health, factors affecting health
13. Community Hygiene: Introduction to Social, Industrial and School hygiene

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

Course Code: ZOO-MDC-2-201
Course Title: UNDERSTANDING HUMAN BODY

Total Hours= 45 (Theory=45) **Credits= 3 (L=3 + P=0)**
Total Marks= 75 (Theory=75)
Theory Marks= 75 {67 (Theory examination) + 08 (Internal assessment)}
Duration for final examination: 3 hours

Note: - This is a three credit introductory course. It is not the same as the courses studied by the students at the 12th level. It does not belong to the core Zoology. It is an interdisciplinary course and can be useful for all the disciplines of Science, Humanities, Commerce, etc. The syllabus can be taught by teachers from any discipline who knows the subject. Therefore, the course may be offered to students of all disciplines irrespective of the subjects studied at the 12th level.

Theory

Learning Outcomes:
The students will be able to:

- *Understand the major organs and systems of the human body.*
- *Explain the physiological processes that support life.*

UNIT 1

Integumentary system
Skin: Structure and functions

Digestive System
Parts and functions of the Alimentary canal, Digestive Glands (Salivary glands, Liver and Pancreas), Basic Physiology of Digestion.

Respiratory System
Parts and functions of the Respiratory tract, Breathing mechanism (Inhalation and Exhalation).

UNIT 2

Cardiovascular System
Structure of Heart, Types of blood vessels (Arteries and Veins), Blood groups, Rh factor, Functions of blood.

Excretory System
Parts and functions of the excretory system.

UNIT 3

Skeletal System
Parts of Axial and Appendicular Skeleton, Ligaments, Tendons, Differences between Bone and Cartilage.

Muscular System
Types of Muscles, Structure of Skeletal Muscle, Functions of the Muscular system.

Nervous System
Major parts of the Brain, Structure of Neuron

UNIT 4

Reproductive System
Parts and functions of the Male and Female reproductive system.

Endocrine System
Major Glands, their hormones and functions (Hypothalamus, Pituitary, Thyroid, Pancreas, Adrenal).

SUGGESTED READINGS

1. Taneja, S.K.: Biochemistry & Animal Physiology, Trueman Book Co., 1997.
2. Guyton, A.S.: Text Book of Medical Physiology, 14th edition, W.B. Saunders Company, 2020.
3. Pradeep's A Text Book of Biology (2024-25) ISBN: 978939564966 Pradeep Publication India
4. Modern's ABC Plus of Biology (Part I & II) Arora&Sabharwal Notebook, 1 January 2021
5. Human Physiology 2007 Stuart Fox (Author) Publisher: McGraw-Hill Education; 10th edition (16 February 2007)

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1.5 marks each. Two questions of 13 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

Course Code: ZOO-MDC-3-301
Course Title: First Aid and Emergency Management

Total Hours= 45 (Theory=45) **Credits= 3 (L=2 + P=1)**
Total Marks= 75 (Theory=75)
Theory Marks= 50 {45 (Theory examination) + 05 (Internal assessment)}
Practical Marks= 25 {22 (Practical examination) + 03 (Internal assessment)}
Duration for final examination: 3 hours

Note: - This is a three credit introductory course. It is not the same as the courses studied by the students at the 12th level. It does not belong to the core Zoology. It is an interdisciplinary course and can be useful for all the disciplines of Science, Humanities, Commerce, etc. The syllabus can be taught by teachers from any discipline who knows the subject. Therefore, the course may be offered to students of all disciplines irrespective of the subjects studied at the 12th level.

Theory

Learning Outcomes:
The students will be able to:

- *Understand the principles of first aid and emergency response.*
- *Learn the skills to manage common injuries and medical emergencies.*
- *Apply basic life support skills until professional help arrives.*

UNIT 1

Introduction to First Aid:
First aid: Meaning, importance of first aid, first aider, laws of first aid, contents of an ideal first aid kit.

UNIT 2

Procedures and Techniques of First Aid:
Dressing, bandaging, splinting, transportation of the injured, CPR (steps in performing CPR)

UNIT 3

First Aid in Emergency:
Drowning, fractures in bone, shock, wound, burns, haemorrhages, bleeding (types and prevention), accidents, poisoning, bites and stings.

UNIT 4

Community Emergencies
First Aid during emergencies like fire explosions, famines, floods, earthquakes, Disaster Management.

Instructions For Paper Setter and candidates

Nine questions are to be set. Question No.1 is compulsory, consisting of short answer type questions covering the whole syllabus. It will have 9 parts of 1 mark each. Two questions of 9 marks each are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

ZOO-MDC-3-301: First Aid and Emergency Management

Practical

1. Assembling the contents of a basic first aid kit.
2. Demonstration of CPR (Using Manikins/Volunteers/Videos)
3. Role-play/Demonstration for medical emergencies (heart attack, fainting)
4. Demonstration of the following through videos/photographs
    - First Aid for Drowning Victim
    - Wound Dressing and Bleeding Control
    - First Aid for Burns
    - First Aid for Bites and Stings

SUGGESTED READINGS

- Natarajan**K. First Aid and Emergency Care,**Jaypee Brothers Medical Publishers.
- RajuS.M. **Textbook of First Aid and Emergency Management,** CBS Publishers & Distributor
- St. John Ambulance (India), **Manual on First Aid and Emergency Care,**The Indian Red Cross Society.
- ClementI. **Textbook on First Aid & Emergency Nursing,**Jaypee Brothers Medical Publishers, 1st Edition (2012). **ISBN:** 9789350259870.
- Gupta H.K. **Disaster Management,** Universities Press, 1st Edition (2003). **ISBN:**9788173714566.

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